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Unlocking the secrets of kangaroo locomotor energetics: Postural adaptations underpin increased tendon stress in hopping kangaroos

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Abstract

Hopping kangaroos exhibit remarkably little change in their rate of metabolic energy expenditure with locomotor speed compared to other running animals. This phenomenon may be related to greater elastic energy savings due to increasing tendon stress; however, the mechanisms which enable the rise in stress remain poorly understood. In this study, we created a three-dimensional (3D) kangaroo musculoskeletal model, integrating 3D motion capture and force plate data, to analyse the kinematics and kinetics of hopping red and grey kangaroos. Using our model, we evaluated how body mass and speed influence (i) hindlimb posture, (ii) effective mechanical advantage (EMA), and (iii) the associated tendon stress in the ankle extensors and (iv) ankle work during hopping. We found that increasing ankle dorsiflexion and metatarsophalangeal plantarflexion likely played an important role in decreasing ankle EMA by altering both the muscle and external moment arms, which subsequently increased energy absorption and peak tendon stress at the ankle. Surprisingly, kangaroo hindlimb posture appeared to contribute to increased tendon stress, thereby elucidating a potential mechanism behind the increase in stress with speed. These posture-mediated increases in elastic energy savings could be a key factor enabling kangaroos to achieve energetic benefits at faster hopping speeds, but may limit the performance of large kangaroos due to the risk of tendon rupture.

eLife assessment

This **valuable** biomechanical analysis of kangaroo kinematics and kinetics across a range of hopping speeds and masses is a step towards understanding a long-standing problem in locomotion biomechanics: the mechanism for how, unlike other mammals, kangaroos are able to increase hopping speed without a concomitant increase in metabolic cost. Based on their suggestion that kangaroo posture changes with speed increase tendon stress/strain and hence elastic energy storage/return, the authors imply (but do not show quantitatively or qualitatively) that the greater tendon elastic energy storage/return counteracts the increased cost of generating muscular force at faster speeds and allows for the invariance in metabolic cost. The methods are impressive, but there is currently only limited evidence for increased tendon stress/strain at faster speeds, and the support for any conclusion metabolic energy expenditure is **inadequate**.

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Introduction

Kangaroos and other macropods are unique in both their morphology and their locomotor style. At slow speeds they use a pentapedal gait, where the forelimbs, the hindlimbs, and the tail all contact the ground, while at faster movement speeds they use their distinctive hopping gait (Dawson and Taylor 1973, O'Connor et al. 2014). Their uniqueness extends into their energetics of locomotion. As far back as the 19th century, researchers noticed that the metabolic cost of running in quadrupeds and bipeds, like dogs, horses and humans, increased linearly with speed (Zuntz 1897, Taylor et al. 1970, Heglund et al. 1982, Taylor et al. 1982). To explain why metabolic rate increased at faster running speeds among quadrupeds and bipeds, Kram and Taylor (1990) refined the 'cost of generating force' hypothesis (Taylor et al. 1980). They reasoned that the decrease in contact time with increased speed, reflects an increase in the rate of generating muscle force, and the rate of cross-bridge cycling (Kram and Taylor 1990). This was supported for a diverse range of running animals, suggesting that metabolic rate was inversely proportional to contact time (Kram and Taylor 1990). Yet hopping macropods appear to defy this trend. On treadmills, both red kangaroos (~20 kg) and tammar wallabies (~5 kg) showed little to no increase in the rate of oxygen consumption with increased hopping speed (Dawson and Taylor 1973, Baudinette et al. 1992, Kram and Dawson 1998). The underlying mechanisms explaining how macropods are able to uncouple hopping speed and energy cost is not completely understood (Thornton et al. 2022).

The ability to uncouple speed and energy expenditure in macropods is related to the behaviour of their ankle extensor muscle-tendon units, which store and return elastic strain energy (Morgan et al. 1978, Biewener et al. 2004b, McGowan et al. 2005). In tammar wallabies, ankle tendon stress increased with hopping speed, leading to a greater rise in elastic strain energy return than muscle work, which increases the proportion of work done by tendon recoil while muscle work remains near constant (Baudinette and Biewener 1998, Biewener et al. 1998). Size-related differences in ankle extensor tendon morphology (Bennett and Taylor 1995, McGowan et al. 2008), and the resultant low strain energy return, may explain why small (<3 kg) hopping macropods and rodents appear not to be afforded the energetic benefits observed in larger macropods (Thompson et al. 1980, Biewener et al. 1981, Biewener et al. 1998) (but see Christensen et al. (2022)). However tendon morphology alone is insufficient to explain why large macropods can increase speed without cost, while large quadrupeds with similar tendon morphology cannot (Dawson and Webster 2010). The most obvious difference between macropods and other mammals is their hopping gait, but previously proposed mechanisms to explain how hopping could reduce the metabolic cost of generating muscle force, such as near-constant stride frequency (Heglund and Taylor 1988, Dawson and Webster 2010) or respiratory-stride coupling (Baudinette et al. 1987), do not distinguish between small and large macropods, nor galloping quadrupeds (McGowan and Collins 2018).

Postural changes are another mechanism which could contribute to reduced energetic costs by altering the leverage of limb muscles. The effective mechanical advantage (EMA) is the ratio of the internal muscle-tendon moment arm (the perpendicular distance between the muscle's force-generating line-of-action and the joint centre) and the ground reaction force (GRF) moment arm (the perpendicular distance between the vector of the GRF to the joint centre). Smaller EMA requires greater muscle force to produce a given force on the ground, thereby demanding a greater volume of active muscle, and presumably higher metabolic rates. In humans, an increase in limb flexion and decrease in limb EMA recruits a greater volume of muscle and increases the metabolic cost of running compared to walking (Biewener et al. 2004a, Kipp et al. 2018, Allen et al. 2022). Elephants— one of the largest and noticeably most upright animals, do not have discrete gait transitions, but rather continuously and substantially decrease EMA in limb joints when moving faster, which has also been linked to the increase in metabolic cost of locomotion with speed (Ren et al. 2010, Langman et al. 2012). EMA has also been observed to change with size in terrestrial mammals. Smaller animals tend to move in crouched postures with limbs becoming progressively more extended as body mass increases (Biewener 1989). Yet, rather than metabolic cost, this postural transition appeared to be driven by the need to reduce the size-related increases in tissue stress (Dick and Clemente 2017, Clemente and Dick 2023). As a consequence of transitioning to more upright limb postures, musculoskeletal stresses in terrestrial mammals are independent of body mass (Biewener 1989, 2005).

Macropods, in contrast, maintain a crouched posture during hopping. Kram and Dawson (1998) explored whether kangaroos transitioned to a more extended limb posture with increasing speed as a potential mechanism for their constant metabolic rate, but did not detect a change in EMA at the ankle across speeds from 4.3 to 9.7 m s^{-1}. Further, ankle posture and EMA appears to vary weakly (McGowan et al. 2008) or not at all with body mass in macropods (Bennett and Taylor 1995, Snelling et al. 2017). As a consequence, stress would be expected to increase with both hopping speed and body mass. High tendon stresses are required for high strain energy return (strain energy ϑ $stress^2$) (Biewener and Baudinette 1995). Given that tendon recoil plays a pivotal role in hopping gaits, it is perhaps unsurprising that tendon stress approaches the safe limit in larger kangaroos. The ankle extensor tendons in a moderately sized male red kangaroo (46.1 kg) operate with safety factors near two even at slow hopping speeds (3.9 m s^{-1}) (Kram and Dawson 1998). Tendon stresses are also unusually high in smaller kangaroos. Juvenile and adult western grey kangaroos, ranging in mass from 5.8 to 70.5 kg, all hop with gastrocnemius and plantaris tendon safety factors less than two (Snelling et al. 2017). High tendon stress may not only be a natural consequence of their crouched posture and tendon morphology, but also adaptively selected. Considering this, kangaroos may adjust their posture to increase tendon stress and its associated elastic energy return. If so, there would likely be systematic variation in kangaroo posture with speed and mass—which is yet to be fully explored.

In this study, we investigate the hindlimb kinematics and kinetics in kangaroos hopping at their preferred speed. Specifically, we explore the relationship between changes in posture, EMA, joint work, and tendon stress across a range of hopping speeds and body masses. To do this, we built a musculoskeletal model of a kangaroo based on empirical imaging and dissection data (**Fig. 1a**). We used the musculoskeletal model to calculate ankle EMA throughout the stride by capturing changes in both muscle moment arm and the GRF moment arm, allowing us to explore their individual contributions. We hypothesised that (i) the distal hindlimb, and the ankle joint in particular, would be more flexed when hopping at faster speeds and larger sizes, and (ii) the change in posture and moment arms would contribute to the increase in tendon stress with speed, and may thereby contribute to energetic savings by increasing the amount of positive and negative work done by the ankle.

Results

Ground reaction forces

The stance begins with a breaking phase (negative horizontal component), followed by a propulsive phase (positive horizontal component). At the transition from negative to positive, the GRF was vertical, and this occurred at 42.9±26.9% of stance. The peak in vertical GRF, however, occurred at 46.4±5.3% of stance (**Fig. 2a,b**).

Larger kangaroos and faster speeds were associated with greater magnitudes of peak GRFs (**Fig. 2a,b**; **Suppl. Table 2**). When peak vertical GRF was normalised to body weight, peak forces ranged from 1.4 multiples of body weight (BW) to 3.7 BW per leg (mean: 2.41±0.396 BW) and increased with speed (**Suppl. Fig. 2a**). Smaller kangaroos may experience disproportionately greater GRFs at faster speeds (**Suppl. Fig. 2b**, **Suppl. Table 2**).

Kinematics and posture

Kangaroo hindlimb kinematics varied with both body mass and speed. Higher masses and faster speeds were associated with more crouched hindlimb postures (**Fig. 3a,c**). Size-related changes in posture occurred throughout stance and were distributed as small changes among all hindlimb joints rather than a large shift in any one joint (**Fig. 3d,f**; **Suppl. Table 3**). The hip, knee and ankle tended toward more flexion, and the metatarsophalangeal (MTP) toward greater plantarflexion in larger kangaroos.

In addition to, and independent of, the change in posture with mass, there were substantial postural changes due to speed (**Suppl. Table 3**). Unlike with mass, speed-related changes occurred predominantly during the braking phase, and the change was concentrated in the ankle and MTP joints, with little to no change in the proximal joints (**Fig. 3c,e,g**; **Suppl. Table 3**). There was a significant decrease in ankle plantarflexion at initial ground contact and increase in dorsiflexion at midstance at faster speeds (**Fig. 3g**; **Suppl. Table 3**). Maximum ankle dorsiflexion occurred at 44.8±4.5% of stance, and tended to occur 3.9±0.7% earlier in stance with each 1 m s^{-1} increase in speed ($P<0.001$). MTP range of motion increased with speed due to an increase in MTP plantarflexion prior to midstance rather than a change in dorsiflexion (**Fig. 3g**).

Effective Mechanical Advantage

Effective mechanical advantage (EMA) decreased as the limb became more crouched. The change in EMA with mass and speed was substantial, particularly in the braking period (**Fig. 2c,d**). We evaluated the change in EMA at midstance, as it was approximately the point in the stride where GRF (and therefore tendon stress) was highest. EMA at midstance decreased with body mass, although we did not detect an effect of speed (**Fig. 2c,d**; **Suppl. Table 5**). However, examining whether the decrease in EMA was caused by a decrease in the muscle moment arm, r, or an increase in the external moment arm, R, revealed a more nuanced picture.

Speed, rather than body mass, was associated with a decrease in r at midstance (**Suppl. Fig. 3a,b**; **Suppl. Table 5**). A kangaroo travelling 1 m s^{-1} faster would decrease r by approximately 4.2%. The muscle moment arm reduced with speed for the full range of speeds we measured (1.99 – 4.48 m s^{-1}), with the maximum r occurring when the ankle was at 114.4±0.8°. Maximum ankle dorsiflexion ranged from 114.5° in the slowest trials, to 75.8° in faster trials (**Suppl. Fig. 3g**). The timing of the local minimum r at midstance coincided with the timing of peak ankle dorsiflexion.

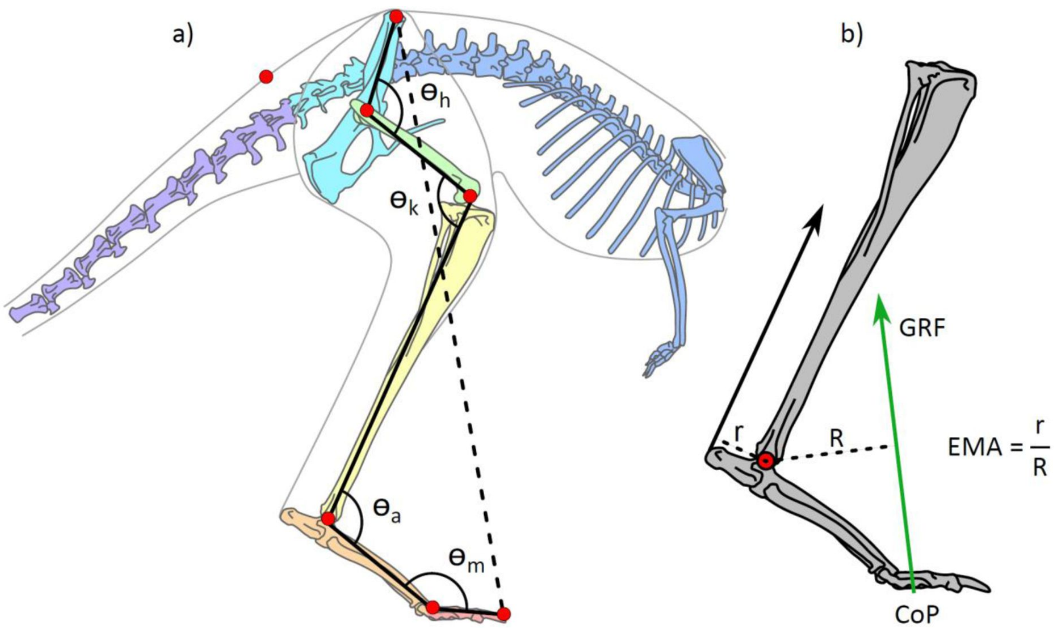

Figure 1

(a) Illustration of the kangaroo model. Total leg length was calculated as the sum of the segment lengths (solid black lines) in the hindlimb and compared to minimum pelvis-to-toe distance (dashed line) to calculate the crouch factor. Joint angles were determined for the hip, h, knee, k, ankle, a, and metatarsophalangeal, m, joints. The model markers (red circles) indicate the position of the reflective markers placed on the kangaroos in the experimental trials and were used to characterize the movement of segments in the musculoskeletal model. (b) Illustration of ankle effective mechanical advantage, EMA, muscle moment arm, r, and external moment arm, R, as the perpendicular distance to the Achilles tendon line of action and ground reaction force (GRF) vector, respectively.

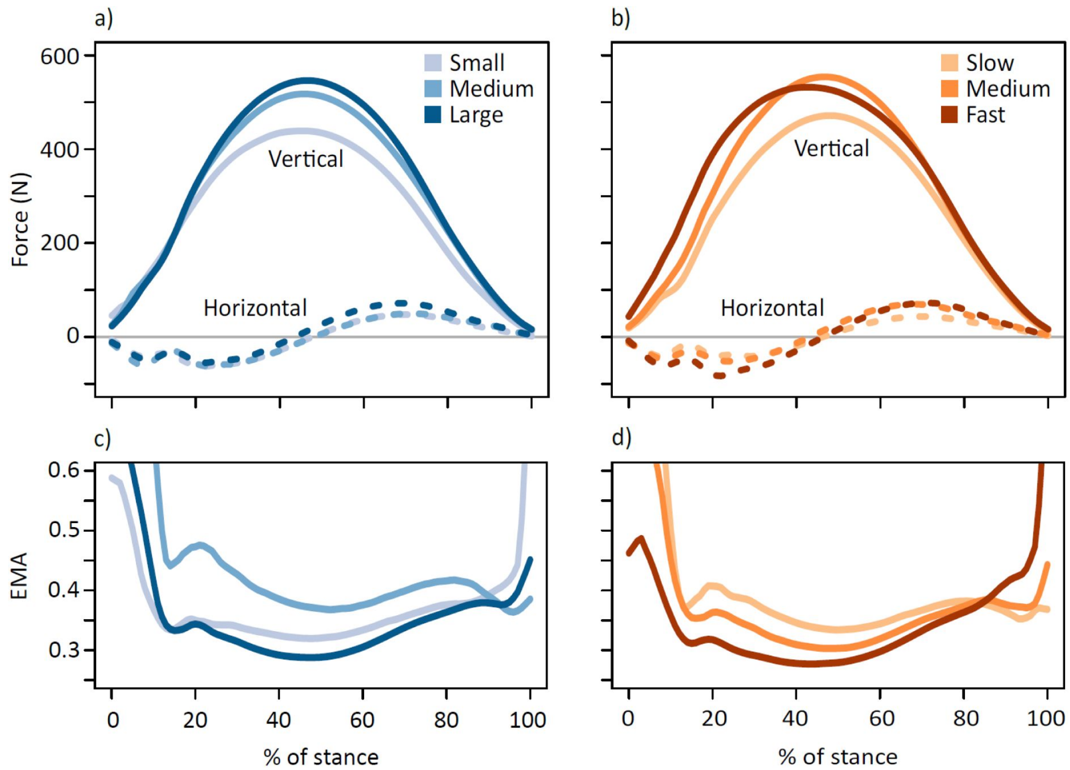

Figure 2

Horizontal (dashed lines) and vertical (solid lines) components of the ground reaction force (GRF) (a) coloured by body mass subsets (small 17.6±2.96 kg, medium 21.5±0.74 kg, large 24.0±1.46 kg) and (b) coloured by speed subsets (slow 2.52±0.25 m s^{-1}, medium 3.11±0.16 m s^{-1}, fast 3.79±0.27 m s^{-1}). In (a) and (b) the medial-lateral component of the GRF is not shown as it remained close to zero, as expected for animals moving in a straight-line path. Lower panels show average time-varying EMA for the ankle joint subset by (c) body mass and (d) speed.

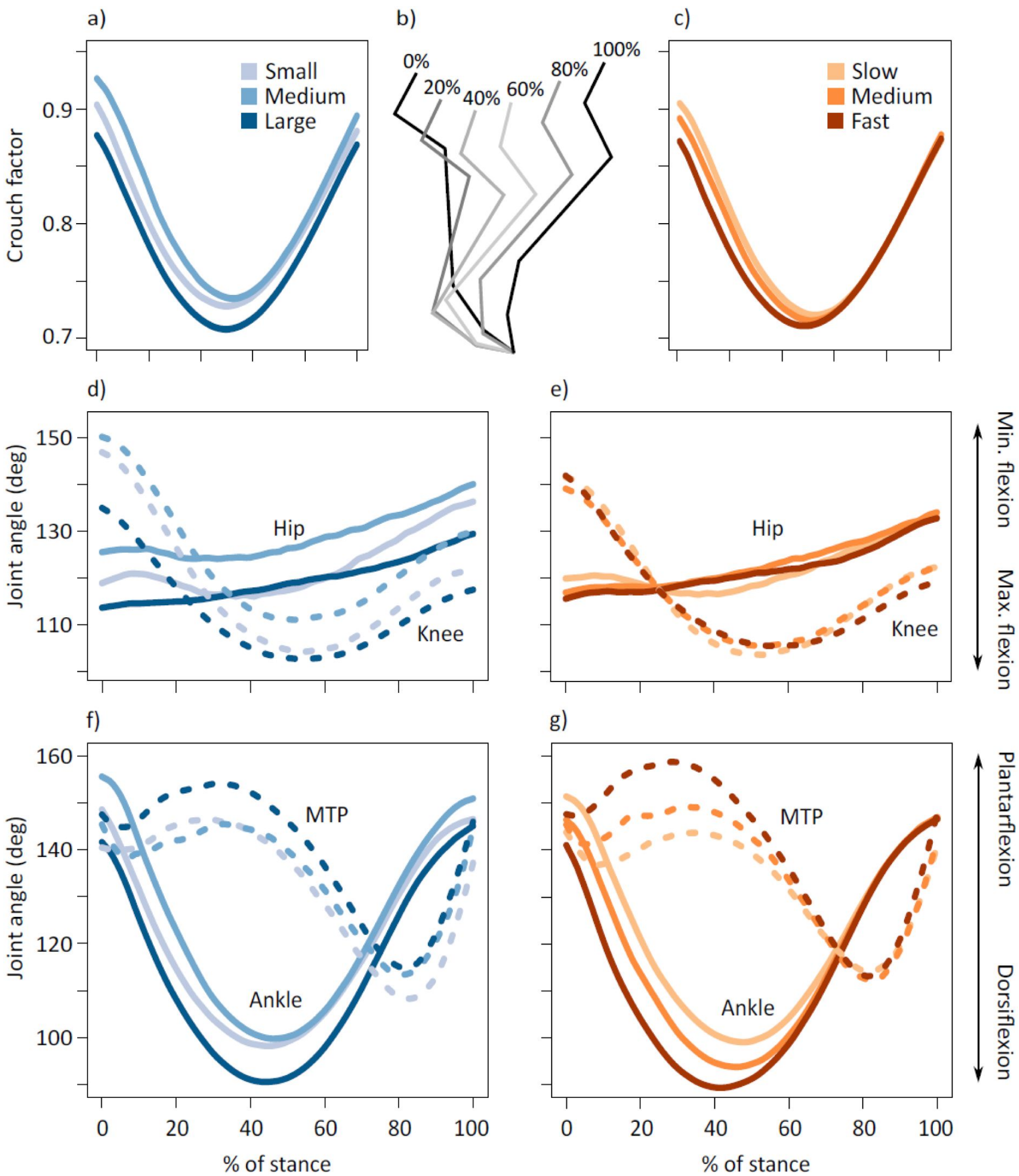

Figure 3

Average time-varying crouch factor of the kangaroo hindlimb grouped by (a) body mass and (c) speed. Position of the limb segments during % stance intervals (b). Average time-varying joint angles for the hip (solid lines) and knee (dashed lines) displayed for kangaroos grouped by (d) body mass and (e) speed. Average time-varying joint angles for the ankle (solid lines) and metatarsophalangeal (MTP) joints (dashed lines) displayed for kangaroos grouped by (f) body mass and (g) speed. Body mass subsets: small 17.6±2.96 kg, medium 21.5±0.74 kg, large 24.0±1.46 kg. Speed subsets: slow 2.52±0.25 m s^{-1}, medium 3.11±0.16 m s^{-1}, fast 3.79±0.27 m s^{-1}.

We observed an increase in R with body mass, but not speed, at midstance (**Suppl. Fig. 3c,d**; **Suppl. Table 5**). We expected R to vary with both mass and speed because the increase in MTP plantarflexion prior to midstance reduced the distance between the ankle joint and the ground, which would increase R since the centre of pressure (CoP) at midstance remained in the same position (**Fig. 6**, **Suppl. Table 2**). Increasing body mass by 4.6 kg or speed by 1 m s^{-1} resulted in a ~20% reduction in ankle height (**Suppl. Fig. 3e,f**). Although, posture alone may not account for the change in R, as the metatarsal segment also increased in length with body mass (1 kg BM corresponded to ~1% longer segment, $P<0.001$, $R^2=0.449$).

Joint moments

The change in the muscle moment arm, r, with speed has further implications for Achilles tendon stress, because force in the tendon is determined by the ratio of the ankle moment to r (**Fig. 6**). As such, the decrease in r with speed would tend to increase tendon force (and thereby tendon stress), as would an increase in ankle moment. We found that speed, rather than mass, was associated with an increase in the maximum ankle moment (**Suppl. Fig. 4**; **Suppl. Table 4**, see supplementary material for details on other hindlimb joint moments).

Tendon stress

Peak tendon stress increased as minimum EMA decreased (**Fig. 7a**). Small decreases in EMA correspond to a nontrivial increase in tendon stress, for instance, reducing EMA from 0.242 (mean minimum EMA of the slow group) to 0.206 (mean minimum EMA of the fast group) was associated with an ~18% increase in tendon stress. Tendon stress increased with body mass and speed (**Suppl. Table 5**). Maximum stress in the ankle extensor tendons occurred at 46.8±4.9% of stance, aligning with the timing of peak vertical GRF and maximum ankle moment. Body mass did not have a significant effect on the timing of peak stress, but peak stress occurred 3.2±0.8% earlier in stance with each 1 m s^{-1} increase in speed ($P<0.001$), reflecting the timing of the peak in ankle dorsiflexion.

Joint work and power

An analysis of joint-level energetics showed the majority of work and power in the hindlimb was performed by the ankle joint (**Fig. 4**; **Suppl. Table 8**). The ankle performed negative work in the braking period, as kinetic and gravitational potential energy was transferred to elastic potential energy by loading the ankle extensor tendons. An increase in negative ankle work was associated with the increase in tendon stress in the ankle extensors ($B=0.750$, $SE=0.074$, $P<0.001$, $R^2=0.511$). Simple linear regression indicated that the increase in negative work was associated with speed but not body mass (**Fig. 5a**; **Suppl. Fig. 5e,f**; **Suppl. Table 6**).

Elastic potential energy was returned to do positive work in the propulsive period as the ankle started extending prior to midstance. Positive ankle work increased with both mass and speed (**Fig. 5a**; **Suppl. Fig. 5e,f**; **Suppl. Table 6**), but since it increased at the same rate as negative work, critically, there was no change in net ankle work with speed (**Fig. 5b**; **Suppl. Fig. 6e,f**; **Suppl. Table 6**).

Discussion

How does posture contribute to kangaroo energetics?

The cost of generating force hypothesis (Taylor et al. 1980) implies that as animals increase locomotor speed and decrease ground contact time, metabolic rate should increase (Kram and Taylor 1990). Macropods defy this trend (Dawson and Taylor 1973, Baudinette et al. 1992, Kram and Dawson 1998). It is likely that the use of their ankle extensor tendons set them apart

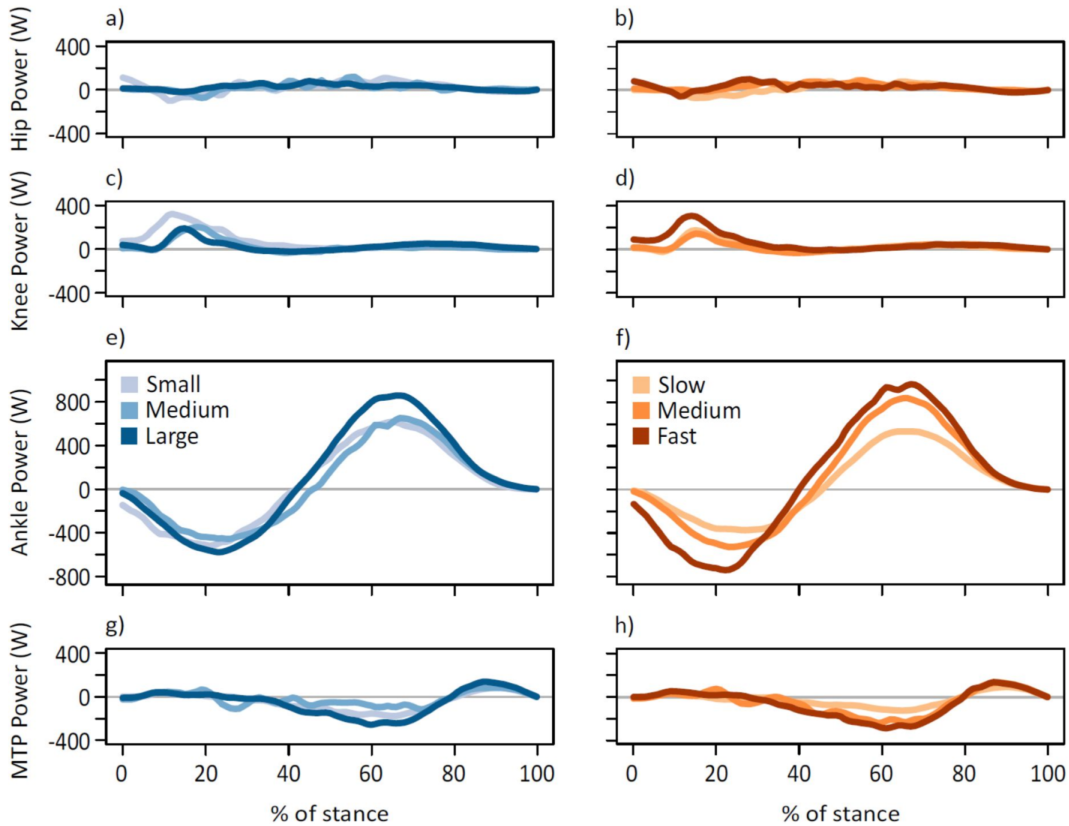

Figure 4

Average time-varying joint powers for the hip (a,b), knee (c, d), ankle (e, f), and MTP (g, h) displayed for kangaroos grouped by body mass (a, c, e, g) and speed (b, d, f, h). Body mass subsets: small 17.6±2.96 kg, medium 21.5±0.74 kg, large 24.0±1.46 kg. Speed subsets: slow 2.52±0.25 m s^{-1}, medium 3.11±0.16 m s^{-1}, fast 3.79±0.27 m s^{-1}.

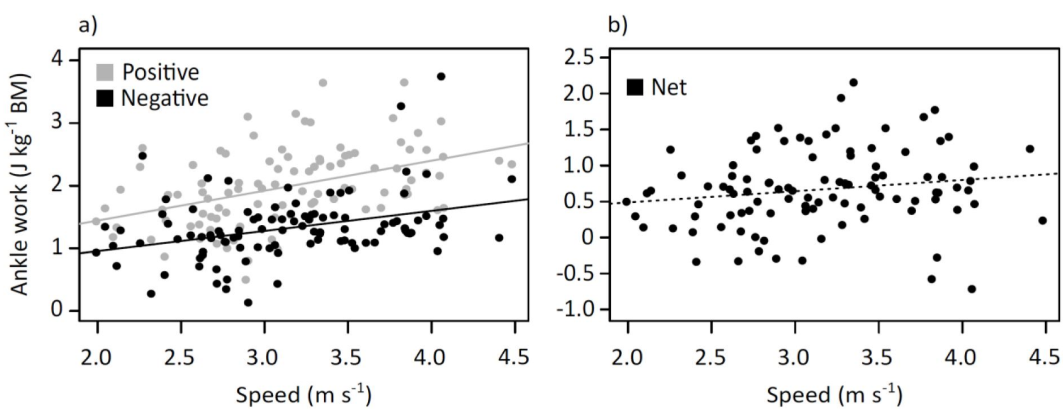

Figure 5

Variation with speed of (a) positive and negative ankle work, and (b) net ankle work.

from other mammals, yet the underlying mechanisms for this were unclear (Bennett and Taylor 1995, Bennett 2000, Thornton et al. 2022). We hypothesised (i) the distal hindlimb would be more flexed when hopping at faster speeds and larger masses, mainly due to the ankle, and (ii) the change in posture and EMA would contribute to the increase in tendon stress.

Tendon stress depends on the muscle moment arm, ankle moment, external moment arm, and ground reaction force (GRF), assuming tendon properties remain unchanged (**Fig. 6**). We detected an increase in peak ankle moment with speed independent of increases in body mass, despite the change in the external moment arm, R, with mass, likely due to a dominant effect of the GRF (ankle moment = GRF · R). Peak GRFs varied with speed as well as mass. Both the increase in ankle moment and a decrease in r with speed tend to increase tendon stress (stress = tendon force / cross-sectional area; tendon force = ankle moment / r). Although the increase in peak GRF with speed explains much of the increase in tendon stress, the significant relationship between tendon stress and ankle EMA suggests that the increase in stress is not solely due to the greater GRFs, but that stress may also be modulated by changes in posture throughout the stride (**Fig. 7a**).

Previously, macropods were reported to have consistent ankle EMA due to the muscle moment arm, r, and the external moment arm, R, scaling similarly with body mass (**Fig. 7b**) (Bennett and Taylor 1995). Our results, however, suggest posture is not as consistent or static as previously reported if speed is taken into account; we found larger and faster kangaroos were more crouched, leading to lower ankle EMA (**Fig. 2c,d**). We were able to explore the individual contributions to ankle EMA using our musculoskeletal model, to demonstrate that changes in the ankle and metatarsophalangeal (MTP) range of motion resulted in changes in r with speed, and R with mass, respectively. These effects were additive; combined, there was a marked decrease in ankle EMA, particularly during the early stance when the ankle extensor tendons are loaded (**Fig. 2c,d**). As such, the change in ankle EMA appears to be a mechanism that contributes to the increase in tendon stress (**Fig. 7a**) and ankle work (**Fig. 5a**).

Kangaroos appear to use changes in posture and EMA to absorb and return energy during stance. A comparison among all the hindlimb joints suggests the ankle was primarily responsible for both the storage and release of work during the stride (**Fig. 4**). Positive work increased with mass and speed, consistent with other studies in other species (Cavagna and Kaneko 1977). However, critically, the amount of negative work absorbed during the stride also increased with speed, and net ankle work did not change with speed, indicating the increase in negative work absorbed matched the increase in positive work that is required to move forward (**Fig. 5b**). Indeed, we observed all trials had a similar positive net ankle work (mean: 0.67 ± 0.54 J kg^{-1}). The consistent net work observed among all speeds suggests the ankle extensors are performing similar amounts of ankle work independent of speed. Previous studies using sonomicrometry have shown that the muscles of tammar wallabies do not shorten considerably during hops, but rather act near-isometrically as a strut (Biewener et al. 1998). Further, these small changes in muscle fibre length did not vary with hopping speed, despite large increases in muscle tendon forces (Biewener et al. 1998). Isometric muscle contractions are efficient (Smith et al. 2005). The cost of generating force hypothesis suggests that faster movement speeds require greater rates of muscle force development, and in turn greater cross bridge cycling rates, driving up metabolic costs (Taylor et al. 1980, Kram and Taylor 1990). The ability for the ankle extensor muscle fibres to remain isometric and produce similar amounts of work at all speeds may help explain why hopping macropods do not follow the energetic trends observed in quadrupedal species. Their ability to do this requires the amount of negative work to increase with speed. Modifying the ankle EMA as speed increases allows an increase in tendon stress (**Fig. 7a**) enabling the storage of more energy during the first half of stance. Since the muscle remains near isometric, no additional work is performed by the muscle, but the overall positive work increases, likely owing to the greater return of elastic strain energy, and enabling faster hopping speeds (**Fig. 6**). Thus,

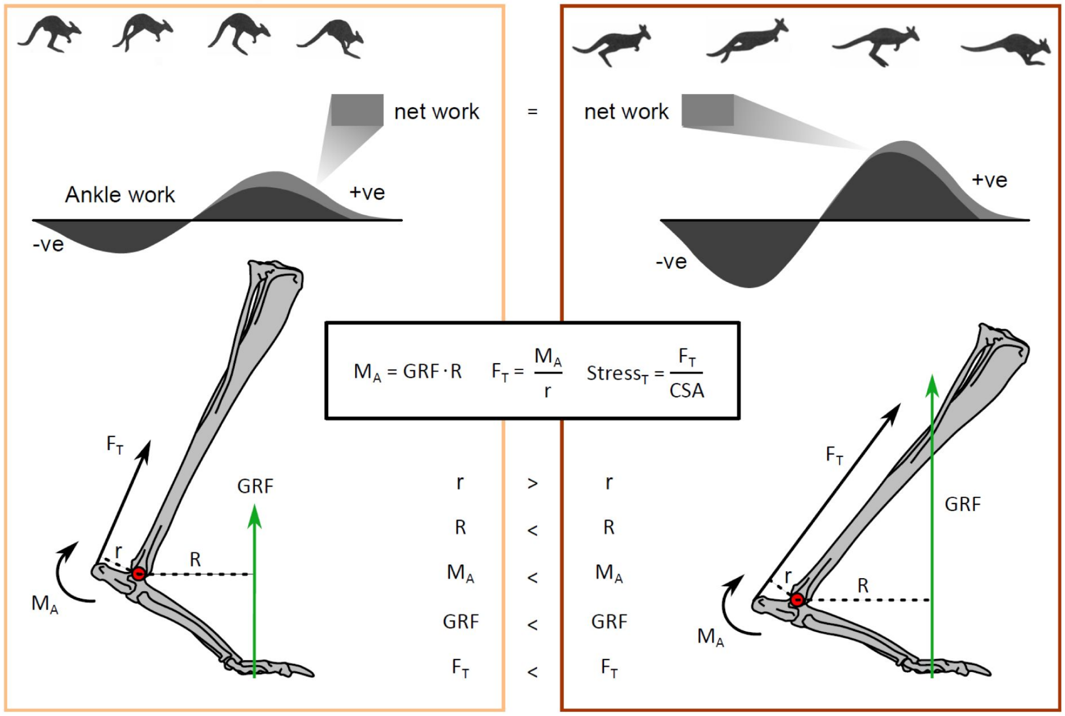

Figure 6

How the relationship between posture and speed is proposed to change tendon stress. Forces are not to scale and joint angles are exaggerated for illustrative clarity. A slow hop (left panel) compared to a fast hop (right panel). The increase in ground reaction force (GRF) with speed, while a more crouched posture changes the muscle moment arm, r, and external moment arm, R, which allows the ankle to do more negative work (storing elastic potential energy in the tendons due to higher tendon stresses), without increasing net work, and thereby metabolic cost.

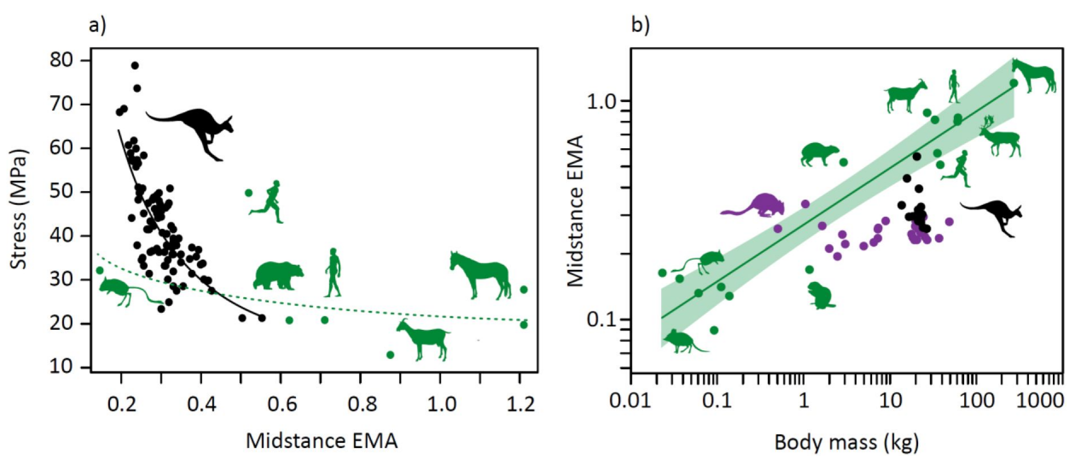

Figure 7

(a) Relationship between ankle effective mechanical advantage, EMA, at midstance and Achilles tendon stress (stress = 11.6 $EMA^{-1.04}$, R^2=0.593) (black), with other mammals (green). (b) Scaling of mean ankle EMA at midstance for each individual kangaroo against body mass (black), with data for a wider range of macropods (purple) (Bennett and Taylor 1995), and other mammals (green, EMA = 0.269 $M^{0.259}$, shaded area 95% confidence interval) (Biewener 1990) shown.

changes in EMA provide a likely mechanism to explain the uncoupling of metabolic energy expenditure with locomotor speed observed in large hopping macropods (Dawson and Taylor 1973, Baudinette et al. 1992, Kram and Dawson 1998).

Why are macropods unique?

No other mammals are known to achieve the same energetic feat as macropods, despite similar tendons or stride parameters (Thornton et al. 2022), but macropods are unique in other ways. Kangaroos operate at much lower EMA values than other mammalian species (**Fig. 7b**). Mammals >18 kg tend to operate with EMA values 0.5 to 1.2. The only mammals with comparable EMA values to kangaroos are rodents <1 kg (**Fig. 7b**) (Biewener 1990), such as kangaroo rats which have relatively thicker tendons that may be less suited for recovering elastic strain energy (Biewener and Blickhan 1988) (but see: Christensen et al. (2022)). **Fig. 7a** shows the non-linear relationship between tendon stress and EMA in kangaroos, quadrupeds and humans. The range of EMA estimates for other mammals suggest they operate in the region of this curve where large changes in EMA would only produce small changes in stress. This implies EMA modulation in other species is not as effective a mechanism to increase tendon stress with increased running speed as that observed in kangaroos.

Were macropods performance or size limited?

The morphology and hopping gait that make kangaroos supremely adapted for efficient locomotion likely also have several performance limitations. One possible consequence is a predicted reduction in manoeuvrability ((Biewener 2005). The high compliance of the tendon would limit the ability to rapidly accelerate, owing to the lag between muscle force production and the transmission of this force to the environment. A second important limitation is the requirement for kangaroos to operate at high tendon stresses. Previous research has suggested that kangaroos locomote at dangerously low safety factors for tendon stress, predicted to be between 1-2 for large kangaroos (Kram and Dawson 1998, McGowan et al. 2008, Snelling et al. 2017, Thornton et al. 2022). Our new insights into the mass and speed modulated changes in EMA suggest that these safety factors may be even lower, likely limiting the maximum body mass that hopping kangaroos can achieve. This suggests that previous projections of tendon stress may have overestimated the body mass at which tendons reach their safety limit ('safety factor' of 1). Snelling et al. (2017) estimated the maximum body mass to remain above this limit was approximately 160 kg, but even if we consider this is a conservative prediction, it is far lower than the projected mass of extinct macropodids (up to 240 kg (Helgen et al. 2006, Janis et al. 2023)). Thus we expect there must be a body mass where postural changes shift from contributing to stress to mitigating it (Dick and Clemente 2017).

This study highlights how EMA may be more dynamic than previously assumed, and how musculoskeletal modelling and simulation approaches can provide insights into direct links between form and function which are often challenging to determine from experiments alone.

Methods

Animals and data collection

Hopping data for red and eastern grey kangaroos was collected at Brisbane's Alma Park Zoo in Queensland, Australia, in accordance with approval by the Ethics and Welfare Committee of the Royal Veterinary College; approval number URN 2010 1051. The dataset includes 16 male and female kangaroos ranging in body mass from 13.7 to 26.6 kg (20.9 ± 3.4 kg). Two juvenile red (*Macropus rufus*) and 11 grey kangaroos (*Macropus giganteus*) were identified, while the remaining three could not be differentiated as either species. Body mass was determined in several ways, primarily by measurements of the kangaroos standing stationary on the force plate.

If the individual did not stop on the force plate, and forward velocity was constant, then body mass was determined by dividing the total impulse across a constant velocity hop cycle (foot strike to foot strike) by the total hop cycle time and further dividing by gravitational acceleration. Finally, if neither of these approaches were sufficient, we interpolated from the relationship between leg marker distances (as proxy for segment lengths) and body mass. These methods produce estimates of body masses close to stationary measurements, where both were available.

Experimental protocol

Kangaroos hopped at their preferred speed down a runway (~10x1.5 m) that was constructed in their enclosure using hessian cloths and stakes (**Suppl. Video 1**). The runway was open at both ends with two force plates (Kistler custom plate (60x60 cm) and AMTI Accugait plate (50x50 cm)) set sequentially in the centre and buried flush with the surface. The force plates recorded ground reaction forces (GRF) in the vertical, horizontal and lateral directions.

A 6-camera 3D motion capture system (Vicon T160 cameras), recorded by Nexus software (Vicon, Oxford, UK) at 200 Hz, was used to record kinematic data. Reflective markers were placed on the animals over the estimated hip, knee, ankle, metatarsophalangeal (MTP) joints; the distal end of phalanx IV; the anterior tip of the ilium; and the base of the tail (**Fig. 1a**). Force plate data was synchronously recorded at 1000 Hz via an analogue to digital board integrated with the Vicon system.

Data analysis

We calculated the stride length from the distance between the ankle or MTP marker coordinates at equivalent time points in the stride. The stance phase was defined as the period when the vertical GRF was greater than 2% of the peak GRF. We determined ground contact duration and total stride duration from the frame rate (200 Hz) and the number of frames from contact to take-off (contact duration) and contact to contact (stride duration), respectively, and calculated the stride frequency. Stride parameter results are detailed in the supplementary material.

In most trials, the stride before and after striking the force plate was visible, providing a total of 173 strides. If two strides were present in a trial, we took the average of the two strides for that trial. Trials were excluded if only one foot landed on the force plate or if the feet did not land near-simultaneously, to give a total of 100 trials. We did not include trials where the kangaroo started from or stopped on the force plate. We assumed GRF was equally shared between each leg and divided the vertical, horizontal and lateral forces in half, and calculated all results for one leg. We normalised GRF by body weight. We assumed the force was applied along phalanx IV and that there was no medial or lateral movement of the centre of pressure (CoP) (**Fig. 1b**). The anterior or posterior movement of the CoP was recorded by the force plate within the motion capture coordinate system.

We calculated the hopping velocity and acceleration of the kangaroo in each trial from the position of the pelvis marker, as this marker was close to the centre of mass and there should have been minimal skin movement artefact. Position data were smoothed using the 'smooth.spline' function in R (version 3.6.3). The average locomotor speed of the trial was taken as the mean horizontal component of the velocity during the aerial phase before and after the stance phase.

Building a kangaroo musculoskeletal model

We created a musculoskeletal model based on the morphology of a kangaroo for use in OpenSim (v3.3; Seth et al. (2018)) (**Fig. 1a**). The skeletal geometry was determined from a computed tomography (CT) scan of a mature western grey kangaroo (*MorphoSource.org*, Duke University, NC, USA). Western grey kangaroos are morphologically similar to eastern grey and red kangaroos (Thornton et al. 2022).

We extracted the skeletal components from the CT scan using Dragonfly (Version 2020.2, Object Research Systems (ORS) Inc., Montreal, Canada) and partitioned the hindlimb into five segments (pelvis, femur, tibia, metatarsals and calcaneus, and phalanges). The segments were imported into Blender (version 3.0.0, *https://www.blender.org/*; Amsterdam, Netherlands) to clean and smooth the bones, align the vertebrae, and export the segments as meshes. We imported the meshes into Rhinoceros (version 6.0, Robert McNeel & Associates, Seattle, WA, USA) to construct the framework of the movement system.

All joints between segments were modelled as hinge joints which constrain motion to a single plane and one degree of freedom (DOF), except the hips which were modelled as ball-and-socket joints with three DOF. The joints were restricted to rotational (no translational) movement. The joints were marked with an origin (joint centre) and coordinate system determined by the movement of the segment (x is abduction/adduction, y is pronation/supination, z is extension/flexion). The limb bones and joints from one leg were mirrored about the sagittal plane to ensure bilateral symmetry.

The segment masses for a base model were determined from measurements of eastern grey and red kangaroos provided in Hopwood (1976). Cylinders set to the length and mass of each segment were used to approximate the segment centre of mass and moments of inertia.

We scaled the model to the size and mass of each kangaroo using the OpenSim scale tool. A static posture was defined based on the 3D positions of the markers at midstance. Each segment was scaled separately, allowing for different scaling factors across segments. The markers with less movement of the skin over the joint (e.g. the MTP marker) were more highly weighted than the markers with substantial skin movement (e.g. the hip and knee markers) in the scaling tool.

Joint kinematics and mechanics

We used inverse kinematics to determine time-varying joint angles during hopping. Inverse kinematics is an optimisation routine which simulates movement by aligning the model markers with the markers in the kinematic data for each time step (**Suppl. Video 2**). We adjusted the weighting on the markers based on the confidence and consistency in the marker position relative to the skeleton. The model movement from inverse kinematics was combined with GRFs in an inverse dynamics analysis to calculate net joint moments for the hip, knee, ankle and MTP joints throughout stance phase. Joint moments were normalised to body weight and leg length.

We calculated instantaneous joint powers for each of the hindlimb joints over the stance phase of hopping as the product of joint moment (not normalised) and joint angular velocity. To determine work, we integrated joint powers with respect to time over discrete periods of positive and negative work, consistent with Dick et al. (2019). For the stance duration of each hop, all periods of positive work were summed and all periods of negative work were summed to determine the positive work, negative work, and net work done at each of the hindlimb joints for a hop cycle. Joint work was normalised to body mass.

Posture and EMA

We evaluated overall hindlimb posture to determine how crouched or upright the hindlimbs were during the stance phase of hopping. The total hindlimb length was determined as the sum of all segment lengths between the joint centres, from the toe to ilium (**Fig. 1a**). The crouch factor (CF) was calculated as the distance between the toe and the ilium marker divided by the total hindlimb length. Larger CF values indicated extended limbs whereas smaller CF values indicated more crouched postures.

We calculated the effective mechanical advantage (EMA) at the ankle as the muscle moment arm of the combined gastrocnemius and plantaris tendon, *r*, divided by the external moment arm, *R* (perpendicular distance between the GRF vector and the ankle joint) (**Fig. 1b**). We dissected (with approval from University of the Sunshine Coast Ethics Committee, ANE2284) a road-killed 27.6 kg male eastern grey kangaroo and used this, combined with published anatomy on the origin and insertion sites, to determine the ankle extensor muscle-tendon unit paths on the skeleton in

OpenSim (Bauschulte 1972, Hopwood and Butterfield 1976, Hopwood and Butterfield 1990). The value *r* to the gastrocnemius and plantaris tendons for all possible ankle angles were determined from OpenSim and scaled to the size of each kangaroo, while *R* was calculated as the perpendicular distance between the ankle marker and the GRF vector at the CoP.

Tendon stress

Ankle extensor tendon forces were estimated as the time-varying ankle moment divided by the time varying Achilles tendon moment arm. The sum of the gastrocnemius and plantaris tendon cross-sectional areas were scaled to kangaroo body mass for each kangaroo by interpolating literature values (Snelling et al. 2017). Forces were divided by tendon cross-sectional area to calculate tendon stress. We excluded the third ankle extensor tendon (flexor digitorum longus), which stores ~10% of strain energy of the ankle extensors in tammar wallabies (Biewener and Baudinette 1995).

Statistics

We used multiple linear regression (lm function in R, v. 3.6.3, Vienna, Austria) to determine the effects of body mass and speed on the stride parameters, ground reaction forces, joint angles and CF, joint moments, joint work and power, and tendon stress. We considered the interaction of body mass and speed first, and removed the interaction term from the linear model if the interaction was not significant. Effects were considered significant at the $p<0.05$ level. Species was not used as a factor in the analysis as there was no systematic difference in outcome measures between kangaroo species.

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Supplementary Results

Stride parameters

Preferred hopping speed ranged from 1.99 to 4.48 m s^{-1}. Larger kangaroos tended to hop at slightly faster speeds (B=0.048, SE=0.018, P=0.009, R^2=0.057), and due to this weak relationship between body mass and speed, both variables were considered in multiple linear regression models to determine their relative effects on the outcome measures (see Table 1).

Faster speeds were associated with a greater magnitude of acceleration in the braking period of the stance phase, i.e. minimum horizontal acceleration and maximum vertical acceleration (**Suppl. Fig. 1a**). There was no significant relationship between body mass and acceleration; however, there was a significant interaction between body mass and speed on maximum vertical acceleration, whereby smaller kangaroos had a greater change in vertical acceleration between slower and faster hopping speeds than larger kangaroos (Table 1).

Body mass and speed had different effects on ground contact duration (Table 1). There was a slight increase in contact duration in larger kangaroos. A stronger, opposing relationship was found with speed, and as speed increased, contact duration decreased. The relatively tight correlation between contact duration and hopping speed (R^2=0.73) could prove useful for predicting speed when contact duration can be accurately measured. In Suppl. Fig. 1b we combined our data with red kangaroo data from Kram and Dawson (1998) to extend the predictive range of both studies.

Larger kangaroos hopped with longer strides and lower frequencies than smaller kangaroos (Table 1), and stride length also increased with speed (**Suppl. Fig. 1c**). We found a significant decrease in stride frequency with mass and an increase with speed, if we did not consider the interaction term (**Suppl. Fig. 1d**). However, a significant interaction between body mass and speed suggests that larger kangaroos relied more on increases in stride frequency to increase hopping speed compared to smaller kangaroos (Table 1).

Ground reaction forces

It is commonly assumed that the GRF is vertical and at a maximum at midstance or 50% of stance (Bennett and Taylor 1995, Kram and Dawson 1998, McGowan et al. 2008, Snelling et al. 2017) but our results suggest it occurred earlier, at 42.9±26.9% of stance, and the wide range suggests such assumptions should be used with caution.

Kinematics and posture

Kangaroos were maximally crouched at midstance, with crouch factor (CF) reaching a minimum at 50.1±4.2% of stance. Crouch factor (CF) at initial ground contact decreased at faster speeds, although the limb was similarly flexed during midstance (P=0.295). Consequently, CF changed less at faster speeds than slower speeds.

Larger kangaroos had lower hip and knee ranges of motion (ROM) compared to smaller kangaroos (**Fig. 3d**; Suppl. Table 3). In the distal hindlimb, ankle ROM increased with body mass, largely owing to an increase in dorsiflexion at midstance (**Fig. 3f**; Suppl. Table 3). The ROM of the MTP joint did not change with body mass; however, there was both an increase in plantarflexion and a decrease in dorsiflexion, resulting in a shift to larger MTP angles with mass (**Fig. 3f**; Suppl. Table 3).

Joint moments

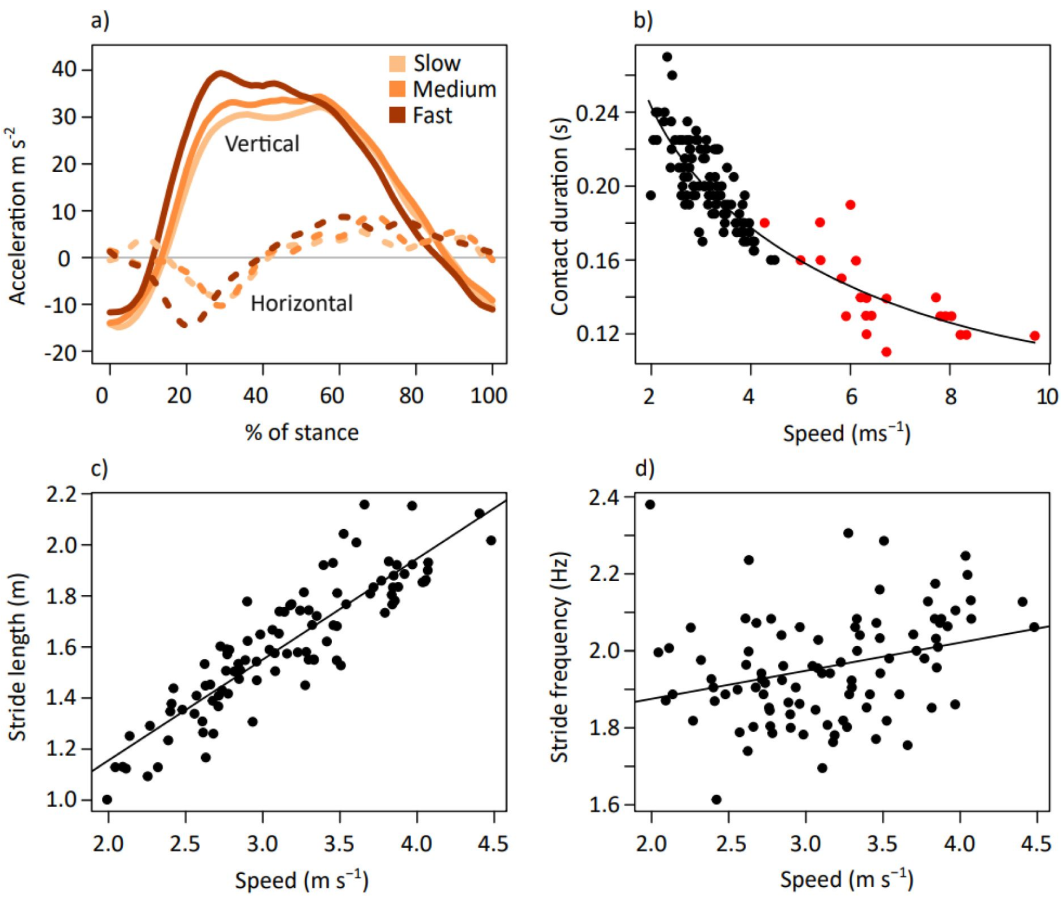
Despite the small changes in joint rotation in the proximal hindlimb, we detected a decrease in magnitude of the dimensionless hip extensor and knee flexor moments with mass (**Suppl. Fig. 4a**, **Suppl. Table 4**), and an increase in magnitude of both joint moments with speed during the braking period of stance (**Suppl. Fig. 4b**). Maximum hip extensor moment and knee flexor moment were significantly influenced by the interaction between body mass and speed, suggesting that larger kangaroos increased the magnitude of the moments at a faster rate with speed compared to smaller kangaroos.

Joint work and power

The MTP was the only joint which did predominantly negative work. The MTP did more negative work with speed, while net MTP work decreased with both mass and speed (**Suppl. Fig. 5g,h**; **Suppl. Fig. 6g,h**; **Suppl. Table 6**). The MTP transition from negative work to positive work at ~80% stance, as the back of the foot started to leave the ground. Conversely, the knee did almost no negative work, and the hip did very little (**Fig. 4**; Suppl. Fig. 5; Suppl. Fig. 6). Net work slightly increased with mass and speed in the hip, while at the knee, net work increased only with speed (**Suppl. Table 6**).

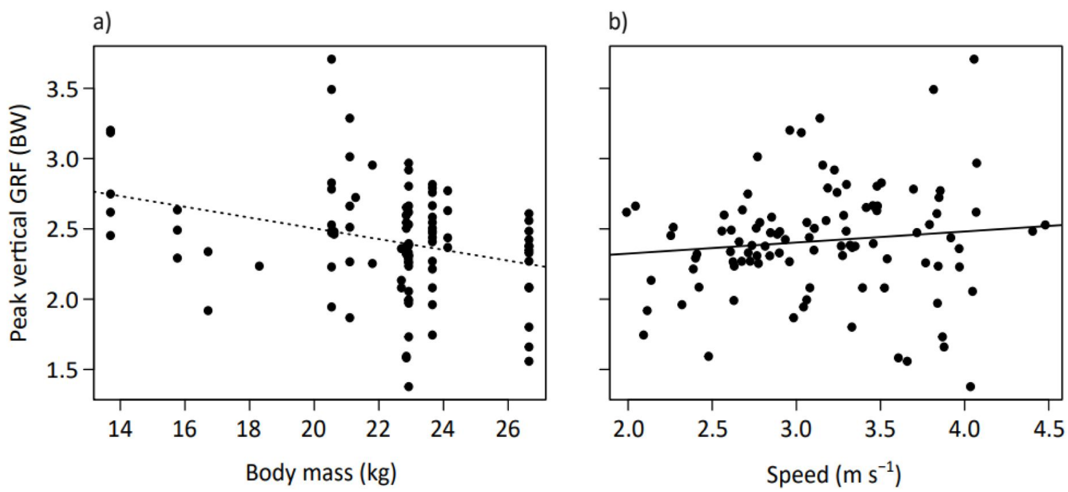
Supplementary Figures

Tables



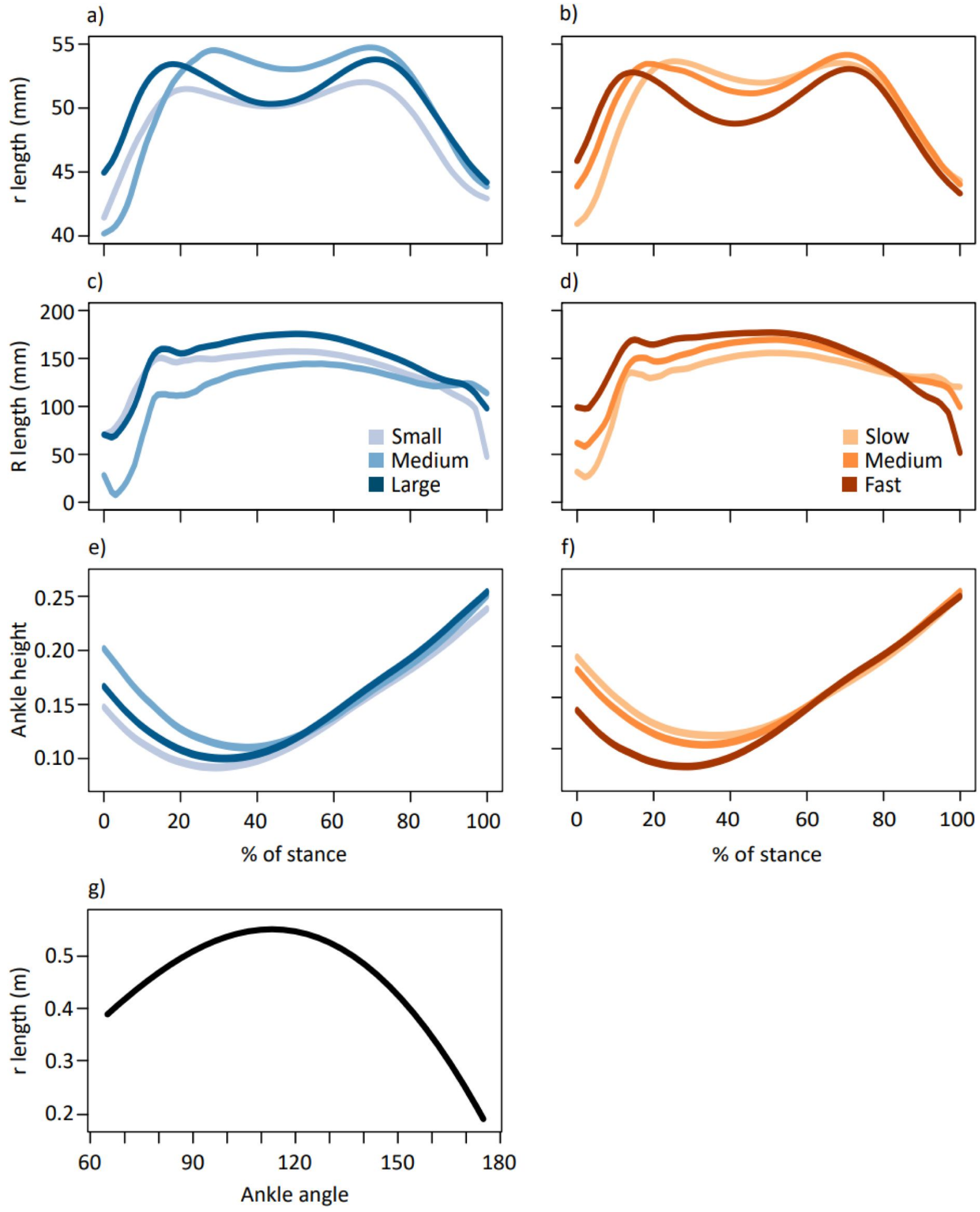
Suppl Figure 1

a) Mean vertical and horizontal components of whole body acceleration for kangaroos in the slow, medium and fast subsets (respectively: 2.52±0.25 m s^{-1}, 3.11±0.16 m s^{-1}, 3.79±0.27 m s^{-1}). (b) Ground contact duration across hopping speeds from current study (black circles) and for red kangaroos reported in Kram and Dawson (1998) (red circles). Regression equation: tc = 0.342speed$^{-0.477}$ where tc is contact duration and s is hopping speed. (c) Relationship between stride length and speed, and (d) stride frequency and speed.



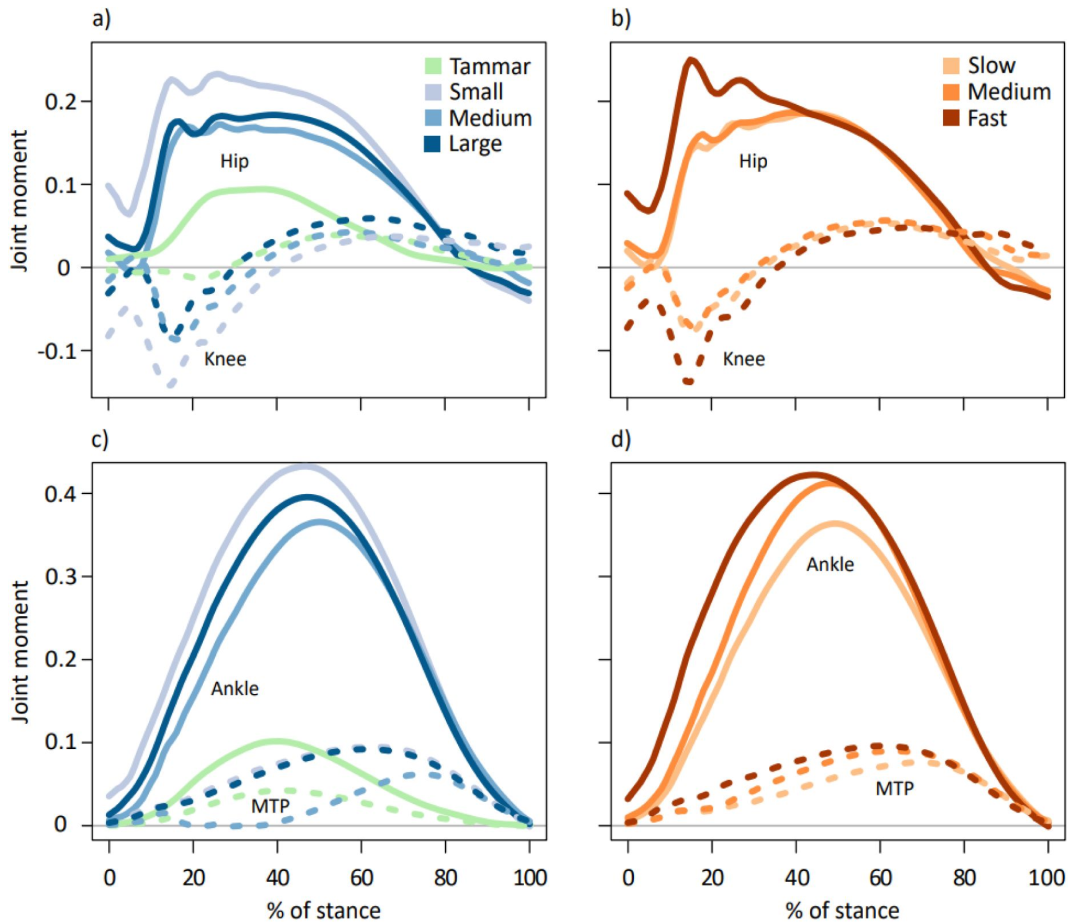
Suppl Figure 2

(a) Relationship between peak vertical GRF as a multiple of body weight (BW) with body mass and (b) with speed. Dotted line is insignificant and solid line is significant, see Table 2 for interaction.



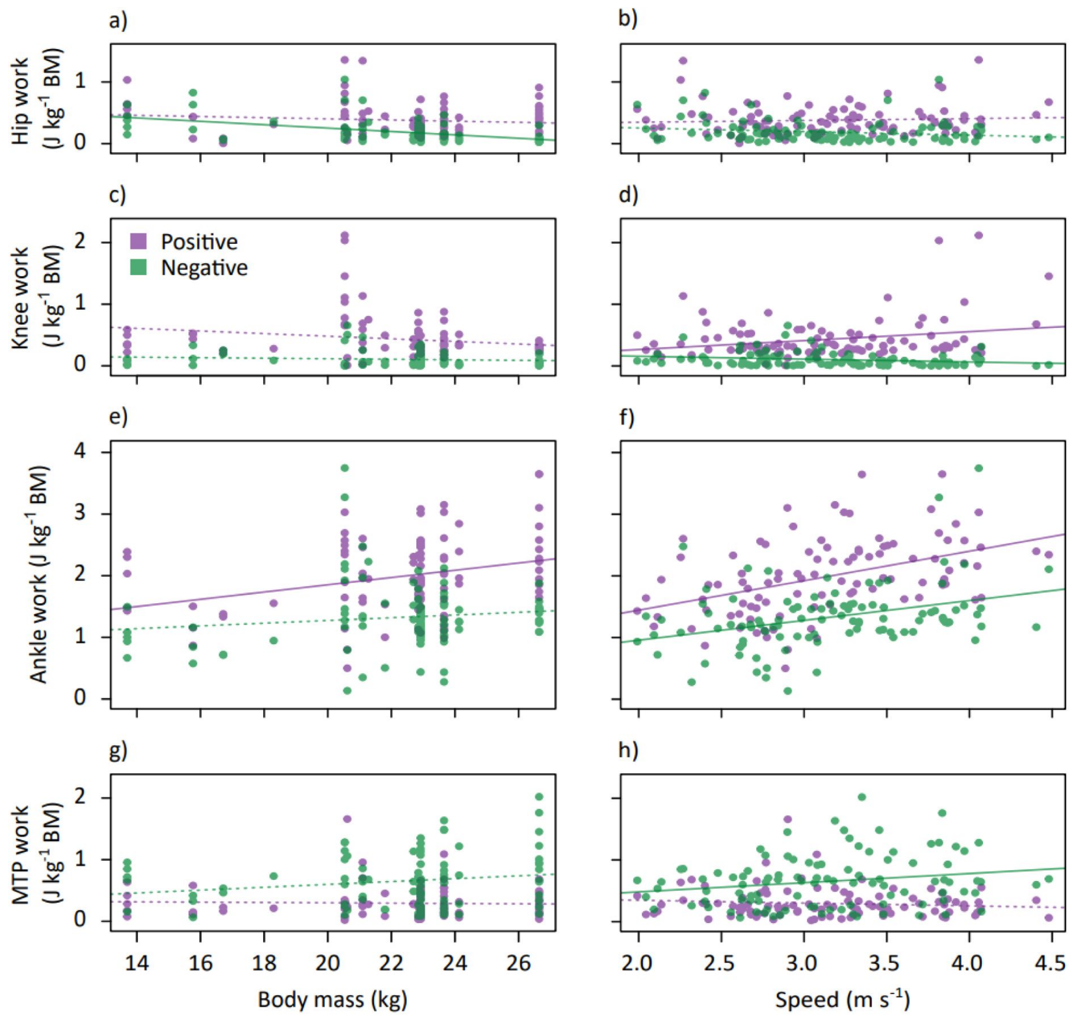
Suppl Figure 3

Average time-varying muscle moment arm, *r*, grouped by (a) body mass and (b) speed; external moment arm, *R*, grouped by (c) body mass and (d) speed. Vertical displacement of the ankle marker from the ground throughout stance grouped by (e) body mass and (f) speed. Ankle angles and corresponding r arm length (g). Body mass subsets: small 17.6±2.96 kg, medium 21.5±0.74 kg, large 24.0±1.46 kg. Speed subsets: slow 2.52±0.25 m s^{-1}, medium 3.11±0.16 m s^{-1}, fast 3.79±0.27 m s^{-1}.



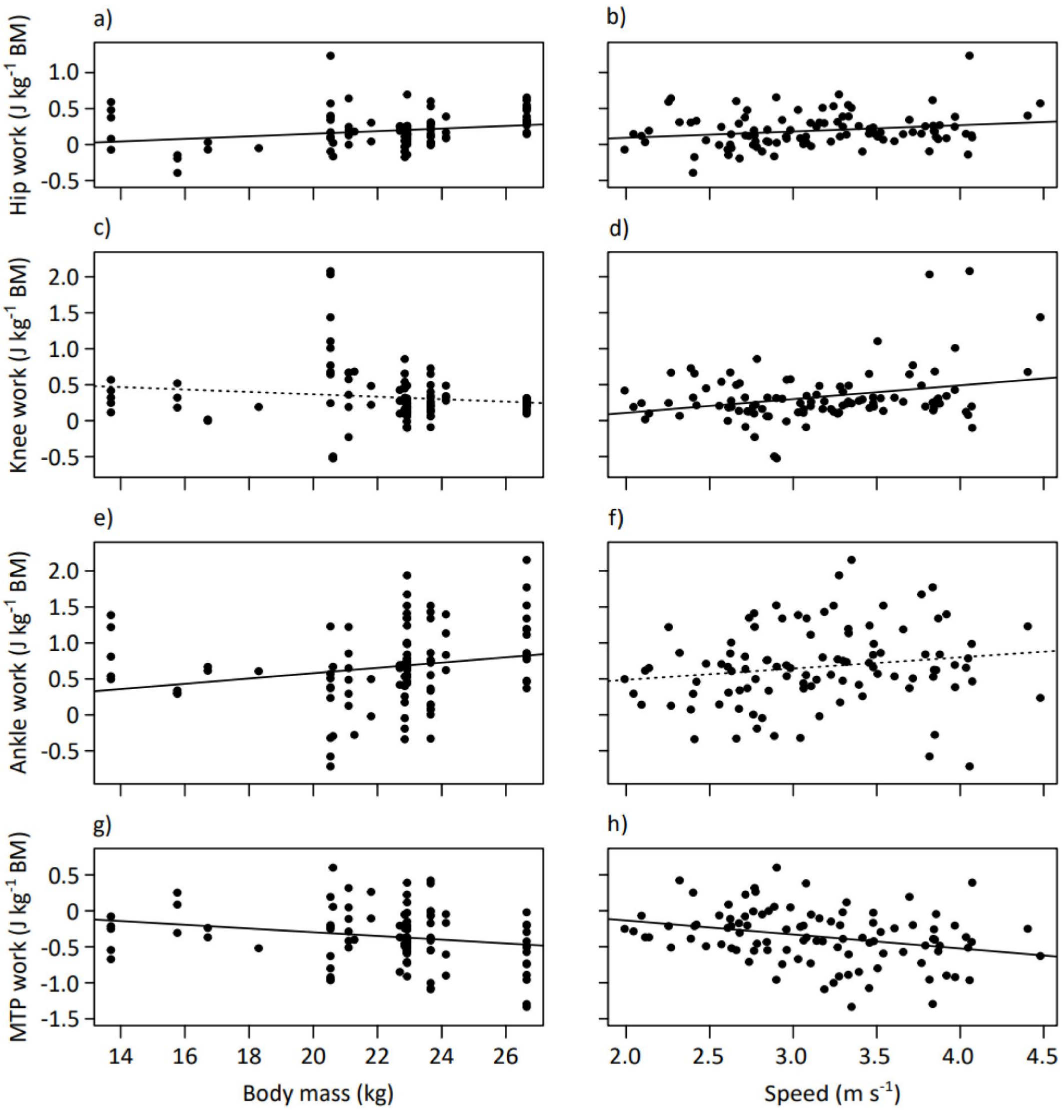
Suppl Figure 4

Average time-varying net joint moments (dimensionless, as moments were divided by body weight * leg length) for the hip (solid lines) and knee (dotted lines) displayed for kangaroos grouped by (a) body mass and (b) speed. Average time-varying net joint moments (dimensionless) for the ankle (solid lines) and metatarsophalangeal (MTP; dotted lines) joints displayed for kangaroos grouped by (c) body mass and (d) speed. Data for tammar wallabies was also included (McGowan et al. 2005) in green. Peak ankle moment occurred at 47.37±4.91 % of the stance phase. Positive values represent extensor moments and negative values represent flexor moments. Body mass subsets: small 17.6±2.96 kg, medium 21.5±0.74 kg, large 24.0±1.46 kg. Speed subsets: slow 2.52±0.25 m s^{-1}, medium 3.11±0.16 m s^{-1}, fast 3.79±0.27 m s^{-1}.



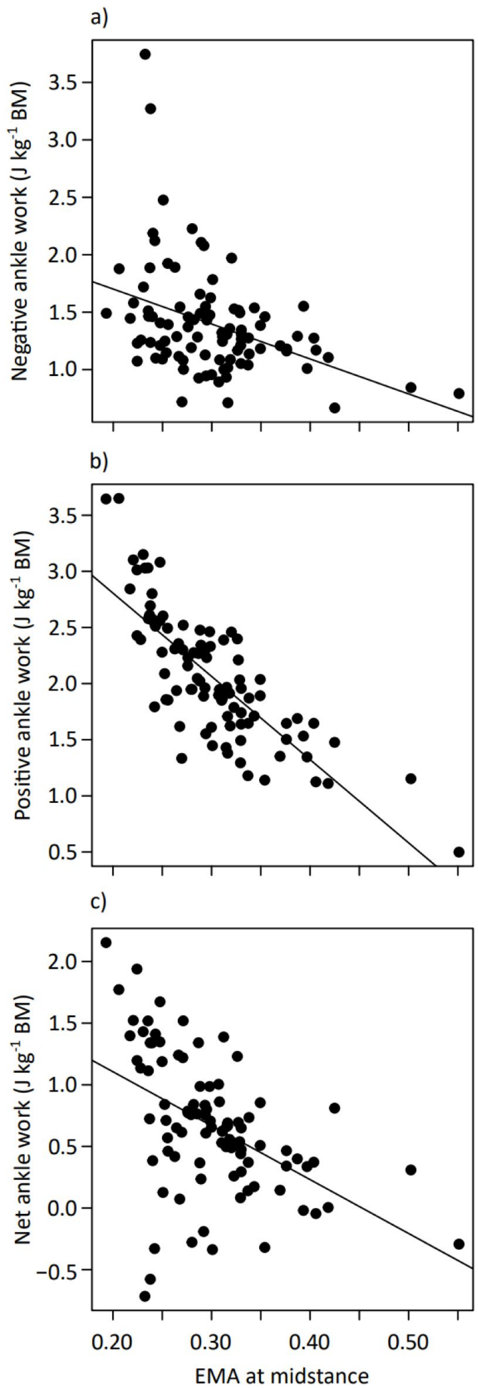
Suppl Figure 5

Positive (purple) and negative (green) joint work over stance for the hip, knee, ankle and MTP plotted against body mass (a,c,e,g) and speed (b,d,f,h). Solid lines represent significant trends, dotted lines are not significant (see **Suppl Table 6**).



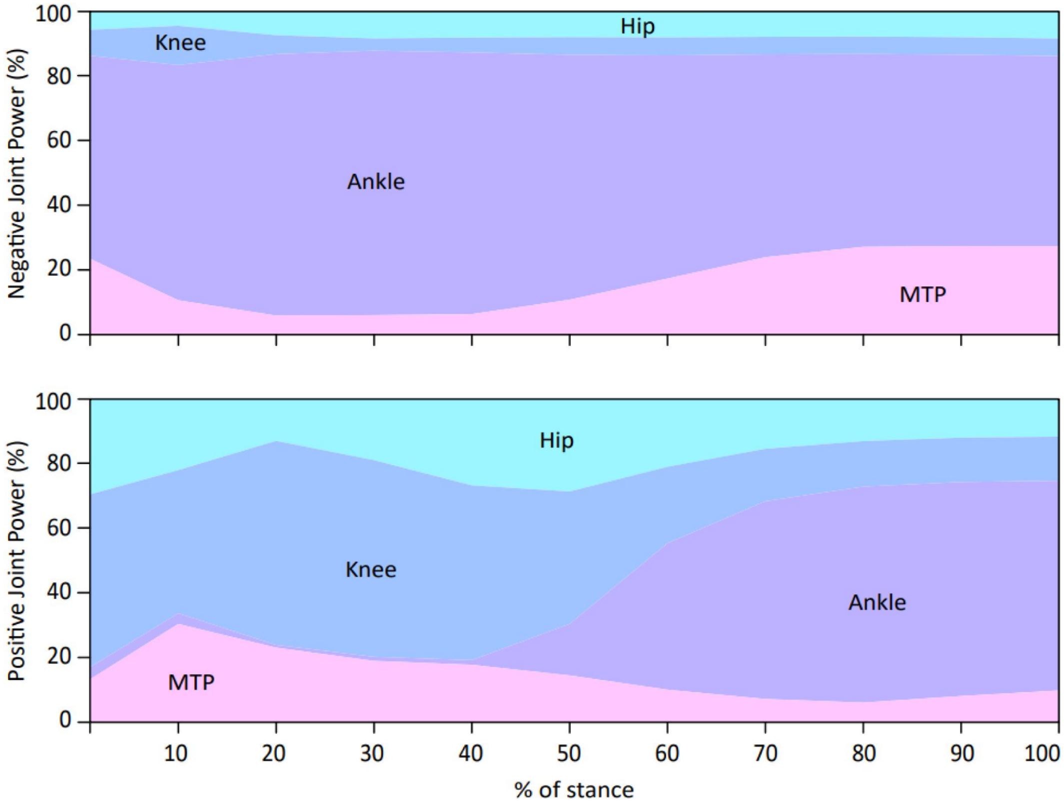
Suppl Figure 6

Net joint work for the hip, knee, ankle and MTP joint over stance plotted against body mass (a,c,e,g) and speed (b,d,f,h). Solid lines represent significant trends, dotted lines are not significant (see **Suppl Table 6**).



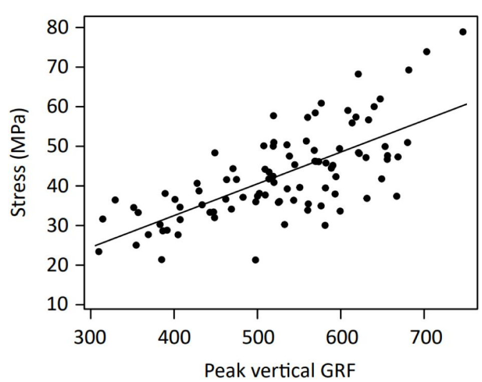
Suppl Figure 7

Negative (a) (B=-3.04, SE=0.75, P<0.001, R^2=0.155), positive (b) (B=-7.42, SE=0.61, P<0.001, R^2=0.622), and net ankle work (c) (B=-4.37, SE=0.84, P<0.001, R^2=0.230) plotted against EMA at 50% of stance.



Suppl Figure 8

Mean instantaneous positive (top panel) and negative (bottom panel) power for each hindlimb joint expressed as a percentage of total limb power (sum of instantaneous joint powers). Means were calculated at 10% stance intervals.



Suppl Figure 9

Peak vertical ground reaction force (GRF) plotted against tendon stress (B=0.080, SE=0.009, P<0.001, R^2=0.486).

	Interaction			Body mass			Speed			R^2
	β	SE	P	β	SE	P	β	SE	P	
Maximum vertical acceleration (m s^{-2})	-1.19	0.567	0.038	2.72	1.64	0.100	32.5	12.5	0.011	0.190
Maximum vertical acceleration (m s^{-2})*				-0.673	0.294	0.024	6.32	1.60	0.000	0.152
Minimum vertical acceleration (m s^{-2})				0.221	0.208	0.290	-0.268	1.13	0.814	0.012
Maximum horizontal acceleration (m s^{-2})				-0.036	0.296	0.903	0.159	1.61	0.922	0.000
Minimum horizontal acceleration (m s^{-2})				0.407	0.370	0.275	-6.05	2.02	0.003	0.087
Contact duration (ms)				2.87	0.393	0.000	-34.2	2.12	0.000	0.735
Stride length (m)				0.014	0.004	0.001	0.376	0.020	0.000	0.815
Stride frequency (Hz)	0.022	0.008	0.007	-0.082	0.023	0.001	-0.392	0.180	0.032	0.303
Stride frequency (Hz)*				-0.019	0.004	0.000	0.102	0.023	0.000	0.245

Suppl Table 1

Stride parameter multiple linear regression results as slopes, standard errors and P-values. Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R^2 and relationships are considered significant at P<0.05.

	Interaction			Body mass			Speed			R^2
	β	SE	P	β	SE	P	β	SE	P	
Normalised peak GRF (BW)	-0.055	0.024	0.026	0.111	0.070	0.117	1.34	0.535	0.014	0.172
Normalised peak GRF (BW)*				-0.045	0.013	0.001	0.142	0.068	0.040	0.128
Peak vertical GRF (N)	-11.5	5.31	0.032	47.9	15.4	0.002	281	117	0.019	0.332
Peak vertical GRF (N)*				15.0	2.76	0.000	28.4	14.9	0.060	0.299
Peak braking GRF (N)	4.87	2.04	0.019	-14.8	5.90	0.014	-130	45.0	0.005	0.218
Peak braking GRF (N)*				-0.917	1.07	0.392	-23.2	5.76	0.000	0.172
Peak propulsive GRF (N)				2.03	0.818	0.015	21.5	4.42	0.000	0.285
CoP location at midstance (mm)				-0.123	1.54	0.937	-15.2	8.34	0.071	0.036
CoP location at midstance corrected for phalanx size (mm)				-0.437	17.0	0.980	-62.7	91.9	0.497	0.005

Suppl Table 2

Ground reaction force and centre of pressure (CoP) multiple linear regression results as slopes, standard errors and P-values. Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R^2 and relationships are considered significant at P<0.05.

	Interaction			Body mass			Speed			R^2
	β	SE	P	β	SE	P	β	SE	P	
Maximum CF				-2.19	1.07	0.043	-15.8	5.74	0.007	0.122
Minimum CF				-2.45	0.885	0.007	-1.53	4.76	0.749	0.064
Change in CF				0.256	0.661	0.699	-14.2	3.55	0.000	0.129
Pelvis pitch ROM (deg)	0.935	0.298	0.002	-2.60	0.862	0.003	-19.9	6.58	0.003	0.100
Pelvis pitch ROM (deg)*				0.066	0.159	0.677	0.534	0.858	0.535	0.008
Hip ROM (deg)				-0.488	0.212	0.024	0.490	1.15	0.670	0.052
Minimum hip flexion (deg)				-0.655	0.315	0.040	0.703	1.70	0.681	0.043
Maximum hip flexion (deg)				-0.167	0.260	0.520	0.213	1.40	0.880	0.004
Knee ROM (deg)	-1.31	0.520	0.013	2.88	1.50	0.058	28.6	11.5	0.014	0.154
Knee ROM (deg)*				-0.850	0.272	0.002	-0.120	1.47	0.935	0.098
Minimum knee flexion (deg)	-1.72	0.649	0.010	3.76	1.88	0.048	39.5	14.3	0.007	0.162
Minimum knee flexion (deg)*				-1.13	0.34	0.001	1.95	1.84	0.294	0.101
Maximum knee flexion (deg)				-0.277	0.265	0.299	2.07	1.43	0.152	0.026
Ankle ROM (deg)				1.02	0.166	0.000	1.63	0.899	0.073	0.339
Peak ankle plantarflexion (deg)				-0.173	0.240	0.474	-3.87	1.30	0.004	0.104
Peak ankle dorsiflexion (deg)				-1.19	0.198	0.000	-5.49	1.07	0.000	0.466
MTP ROM (deg)				-0.098	0.259	0.707	7.16	1.40	0.000	0.219
Peak MTP plantarflexion (deg)				0.875	0.306	0.005	5.64	1.65	0.001	0.216
Peak MTP dorsiflexion (deg)				0.973	0.221	0.000	-1.52	1.20	0.205	0.166

Suppl Table 3

Crouch factor (CF) and kinematics multiple linear regression results as slopes, standard errors and P-values. Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R^2 and relationships are considered significant at P<0.05.

	Interaction			Body mass			Speed			R^2
	β	SE	P	β	SE	P	β	SE	P	
Hip extensor moment	-0.020	0.006	0.001	0.049	0.018	0.007	0.525	0.136	0.000	0.268
Hip extensor moment*				-0.009	0.003	0.010	0.079	0.018	0.000	0.184
Knee extensor moment				0.000	0.001	0.766	0.003	0.008	0.721	0.003
Knee flexor moment	0.020	0.005	0.000	-0.048	0.015	0.002	-0.487	0.116	0.000	0.264
Knee flexor moment*				0.008	0.003	0.007	-0.060	0.015	0.000	0.158
Ankle extensor moment				-0.004	0.004	0.256	0.043	0.020	0.036	0.048
MTP extensor moment				-0.001	0.002	0.767	0.000	0.013	0.986	0.001

Suppl Table 4

Multiple linear regression results of dimensionless peak joint moments as slopes, standard errors and P-values. Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R^2 and relationships are considered significant at P<0.05.

	Interaction			Body mass			Speed			R^2
	β	SE	P	β	SE	P	β	SE	P	
r at midstance (mm)				-0.063	0.088	0.477	-1.88	0.474	0.000	0.173
R at midstance (mm)				3.49	1.03	0.001	-0.93	5.59	0.869	0.117
EMA at midstance				-6.64	1.98	0.001	-11.0	10.7	0.310	0.142
Peak tendon stress (MPa)				1.03	0.385	0.008	5.61	2.08	0.008	0.168
Normalised peak tendon stress				-0.048	0.018	0.008	0.258	0.096	0.008	0.107
Peak tendon stress timing %stance				-0.151	0.154	0.328	-3.13	0.831	0.000	0.159
Ankle height from ground (mm)				-0.298	0.706	0.674	-23.0	3.81	0.000	0.295

Suppl Table 5

Tendon stress and EMA multiple linear regression results as slopes, standard errors and P-values. Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R^2 and relationships are considered significant at $P<0.05$.

	Body mass				Speed			
	β	SE	P	R^2	β	SE	P	R^2
Hip pos	-0.010	0.008	0.238	0.014	0.030	0.044	0.497	0.005
Hip neg	-0.027	0.005	0.000	0.225	-0.058	0.031	0.066	0.034
Hip net	0.018	0.007	0.018	0.055	0.087	0.041	0.034	0.045
Knee pos	-0.021	0.011	0.052	0.038	0.145	0.057	0.013	0.061
Knee neg	-0.004	0.004	0.267	0.013	-0.046	0.020	0.026	0.050
Knee net	-0.017	0.012	0.169	0.019	0.191	0.064	0.003	0.084
Ankle pos	0.059	0.019	0.003	0.087	0.478	0.097	0.000	0.197
Ankle neg	0.022	0.017	0.200	0.017	0.321	0.087	0.000	0.122
Ankle net	0.037	0.017	0.037	0.044	0.156	0.095	0.102	0.027
MTP pos	-0.003	0.008	0.743	0.001	-0.046	0.043	0.286	0.012
MTP neg	0.023	0.014	0.089	0.029	0.149	0.073	0.044	0.041
MTP net	-0.026	0.012	0.035	0.045	-0.194	0.064	0.003	0.086

Suppl Table 6

Joint net positive, negative and net work simple linear regression (lm(joint work ~ mass), lm(joint work ~ speed)) results as slopes, standard errors and P-values. Work is normalised by body mass (BM). The fit of the model is represented by R^2 and relationships are considered significant at $P<0.05$.

Joint work (J kg^{-1} BM)	Interaction			Body mass			Speed			R^2
	β	SE	P	β	SE	P	β	SE	P	
Hip positive				0.046	0.045	0.307	-0.012	0.008	0.161	0.025
Hip negative				-0.021	0.029	0.472	-0.026	0.005	0.000	0.229
Hip net				0.067	0.042	0.110	0.015	0.008	0.058	0.080
Knee positive	-0.072	0.020	0.000	1.766	0.431	0.000	0.175	0.057	0.003	0.241
Knee positive*				0.187	0.057	0.002	-0.030	0.011	0.006	0.133
Knee negative				-0.043	0.021	0.045	-0.002	0.004	0.570	0.053
Knee net	-0.073	0.022	0.002	1.818	0.491	0.000	0.179	0.064	0.007	0.219
Knee net*				0.230	0.064	0.001	-0.028	0.012	0.021	0.133
Ankle positive				0.424	0.099	0.000	0.039	0.018	0.038	0.232
Ankle negative	-0.080	0.032	0.015	2.052	0.707	0.005	0.234	0.093	0.013	0.176
Ankle negative*				0.311	0.091	0.001	0.007	0.017	0.671	0.123
Ankle net	0.122	0.033	0.000	-2.551	0.732	0.001	-0.315	0.096	0.001	0.173
Ankle net*				0.112	0.097	0.250	0.031	0.018	0.084	0.057
MTP positive				-0.045	0.044	0.312	0.000	0.008	0.956	0.012
MTP negative				0.125	0.075	0.101	0.017	0.014	0.217	0.056
MTP net				-0.170	0.066	0.011	-0.018	0.012	0.149	0.106

Suppl Table 7

Positive, negative and net joint work multiple linear regression results as slopes, standard errors and P-values. Work is normalised by body mass (BM). Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R^2 and relationships are considered significant at $P<0.05$.

Joint work (J kg^{-1} BM)	Mean	SD
Hip positive	0.38	0.246
Hip negative	0.187	0.178
Hip net	0.193	0.235
Knee positive	0.432	0.334
Knee negative	0.104	0.117
Knee net	0.328	0.375
Ankle positive	1.99	0.613
Ankle negative	1.324	0.525
Ankle net	0.666	0.542
MTP positive	0.294	0.242
MTP negative	0.652	0.42
MTP net	-0.358	0.377

Suppl Table 8

**The mean and standard deviation of joint work for all trials.
Positive, negative and net work is presented for each joint.**

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Reviewer #1 (Public Review):

Summary:

The study explored the biomechanics of kangaroo hopping across both speed and animal size to try and explain the unique and remarkable energetics of kangaroo locomotion.

Strengths:

The study brings kangaroo locomotion biomechanics into the 21st century. It is a remarkably difficult project to accomplish. There is excellent attention to detail, supported by clear writing and figures.

Weaknesses:

The authors oversell their findings, but the mystery still persists. The manuscript lacks a big-picture summary with pointers to how one might resolve the big question.

General Comments

This is a very impressive tour de force by an all-star collaborative team of researchers. The study represents a tremendous leap forward (pun intended) in terms of our understanding of kangaroo locomotion. Some might wonder why such an unusual species is of much interest. But, in my opinion, the classic study by Dawson and Taylor in 1973 of kangaroos launched the modern era of running biomechanics/energetics and applies to varying degrees to all animals that use bouncing gaits (running, trotting, galloping and of course hopping). The puzzling metabolic energetics findings of Dawson & Taylor (little if any increase in metabolic power despite increasing forward speed) remain a giant unsolved problem in comparative locomotor biomechanics and energetics. It is our "dark matter problem".

This study is certainly a hop towards solving the problem. But, the title of the paper overpromises and the authors present little attempt to provide an overview of the remaining big issues. The study clearly shows that the ankle and to a lesser extent the mtp joint are where the action is. They clearly show in great detail by how much and by what means the ankle joint tendons experience increased stress at faster forward speeds. Since these were zoo animals, direct measures were not feasible, but the conclusion that the tendons are storing and returning more elastic energy per hop at faster speeds is solid. The conclusion that net muscle work per hop changes little from slow to fast forward speeds is also solid. Doing less muscle work can only be good if one is trying to minimize metabolic energy consumption. However, to achieve greater tendon stresses, there must be greater muscle forces. Unless one is willing to reject the premise of the cost of generating force hypothesis, that is an important issue to confront. Further, the present data support the Kram & Dawson finding of decreased contact times at faster forward speeds. Kram & Taylor and subsequent applications of (and challenges to) their approach supports the idea that shorter contact times (tc) require recruiting more expensive muscle fibers and hence greater metabolic costs. Therefore, I think that it is incumbent on the present authors to clarify that this study has still not tied up the metabolic energetics across speed problems and placed a bow atop the package. Fortunately, I am confident that the impressive collective brain power that comprises this author list can craft a paragraph or two that summarizes these ideas and points out how the group is now uniquely and enviably poised to explore the problem more using a dynamic SIMM model that incorporates muscle energetics (perhaps ala' Umberger et al.). Or perhaps they have other ideas about how they can really solve the problem.

I have a few issues with the other half of this study (i.e. animal size effects). I would enjoy reading a new paragraph by these authors in the Discussion that considers the evolutionary origins and implications of such small safety factors. Surely, it would need to be speculative, but that's OK.

https://doi.org/10.7554/eLife.96437.1.sa2

Reviewer #2 (Public Review):

Summary

This is a fascinating topic that has intrigued scientists for decades. I applaud the authors for trying to tackle this enigma. In this manuscript, the authors primarily measured hopping biomechanics data from kangaroos and performed inverse dynamics. While these biomechanical analyses were thorough and impressively incorporated collected anatomical data and an Opensim model, I'm afraid that they did not satisfactorily address how kangaroos can hop faster and not consume more metabolic energy, unique from other animals. Noticeably, the authors did not collect metabolic data nor did they model metabolic rates using their modelling framework. Instead, they performed a somewhat traditional inverse dynamics analysis from multiple animals hopping at a self-selected speed. Within these analyses, the authors largely focused on ankle EMA, discussing its potential importance (because it affects tendon stress, which affects tendon strain energy, which affects muscle mechanics) on the metabolic cost of hopping. However, EMA was roughly estimated (CoP was fixed to the foot, not measured) and did not detectibly associate with hopping speed (see results). Yet, the authors interpret their EMA findings as though it systematically related with speed to explain their theory on how metabolic cost is unique in kangaroos vs. other animals. These speed vs. biomechanics relationships were limited by comparisons across different animals hopping at different speeds and could have been strengthened using repeated measures design. There are also multiple inconsistencies between the authors' theory on how mechanics affect energetics and the cited literature, which leaves me somewhat confused and wanting more clarification and information on how mechanics and energetics relate. My apologies for the less-than-favorable review, I think that this is a neat biomechanics study - but am unsure if it adds much to the literature on the topic of kangaroo hopping energetics in its current form.

https://doi.org/10.7554/eLife.96437.1.sa1

Reviewer #3 (Public Review):

Summary:

The goal of this study is to understand how, unlike other mammals, kangaroos are able to increase hopping speed without a concomitant increase in metabolic cost. They use a biomechancial analysis of kangaroo hopping data across a range of speeds to investigate how posture, effective mechanical advantage, and tendon stress vary with speed and mass. The main finding is that a change in posture leads to increasing effective mechanical advantage with speed, which ultimately increases tendon elastic energy storage and returns via greater tendon strain. Thus kangaroos may be able to conserve energy with increasing speed by flexing more, which increases tendon strain.

Strengths:

The approach and effort invested into collecting this valuable dataset of kangaroo locomotion is impressive. The dataset alone is a valuable contribution.

Weaknesses:

Despite these strengths, I have concerns regarding the strength of the results and the overall clarity of the paper and methods used (which likely influences how convincingly the main results come across).

(1) The paper seems to hinge on the finding that EMA decreases with increasing speed and that this contributes significantly to greater tendon strain estimated with increasing speed. It is very difficult to be convinced by this result for a number of reasons:
• It appears that kangaroos hopped at their preferred speed. Thus the variability observed is across individuals not within. Is this large enough of a range (either within or across subjects) to make conclusions about the effect of speed, without results being susceptible to differences between subjects? In the literature cited, what was the range of speeds measured, and was it within or between subjects?
• Assuming that there is a compelling relationship between EMA and velocity, how reasonable is it to extrapolate to the conclusion that this increases tendon strain and ultimately saves metabolic cost? They correlate EMA with tendon strain, but this would still not suggest a causal relationship (incidentally the p-value for the correlation is not reported). Tendon strain could be increasing with ground reaction force, independent of EMA. Even if there is a correlation between strain and EMA, is it not a mathematical necessity in their model that all else being equal, tendon stress will increase as ema decreases? I may be missing something, but nonetheless, it would be helpful for the authors to clarify the strength of the evidence supporting their conclusions.
• The statistical approach is not well-described. It is not clear what the form of the statistical model used was and whether the analysis treated each trial individually or grouped trials by the kangaroo. There is also no mention of how many trials per kangaroo, or the range of speeds (or masses) tested. Related to this, there is no mention of how different speeds were obtained. It seems that kangaroos hopped at a self-selected pace, thus it appears that not much variation was observed. I appreciate the difficulty of conducting these experiments in a controlled manner, but this doesn't exempt the authors from providing the details of their approach.
• Some figures (Figure 2 for example) present means for one of three speeds, yet the speeds are not reported (except in the legend) nor how these bins were determined, nor how many trials or kangaroos fit in each bin. A similar comment applies to the mass categories. It would be more convincing if the authors plotted the main metrics vs. speed to illustrate the significant trends they are reporting.

(2) The significance of the effects of mass is not clear. The introduction and abstract suggest that the paper is focused on the effect of speed, yet the effects of mass are reported throughout as well, without a clear understanding of the significance. This weakness is further exaggerated by the fact that the details of the subject masses are not reported.

(3) The paper needs to be significantly re-written to better incorporate the methods into the results section. Since the results come before the methods, some of the methods must necessarily be described such that the study can be understood at some level without turning to the dedicated methods section. As written, it is very difficult to understand the basis of the approach, analysis, and metrics without turning to the methods.

https://doi.org/10.7554/eLife.96437.1.sa0

Author response:

We thank the reviewers for their overall positive assessments and constructive feedback

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Thank you for the kind words

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Thank you for the kind words

> *This study is certainly a hop towards solving the problem. But, the title of the paper overpromises and the authors present little attempt to provide an overview of the remaining big issues.*

We will modify the title to reflect this comment.

> *The study clearly shows that the ankle and to a lesser extent the mtp joint are where the action is. They clearly show in great detail by how much and by what means the ankle joint tendons experience increased stress at faster forward speeds.*
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> *Since these were zoo animals, direct measures were not feasible, but the conclusion that the tendons are storing and returning more elastic energy per hop at faster speeds is solid.*
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You have raised important points, thank you for this feedback. We will add a paragraph discussing the limitations of our study and ensure the revised manuscript makes it clear which mysteries remain. We intend to address muscle forces, contact time, and energetics in future work when we have implemented all hindlimb muscles within the musculoskeletal model.

> *I have a few issues with the other half of this study (i.e. animal size effects). I would enjoy reading a new paragraph by these authors in the Discussion that considers the evolutionary origins and implications of such small safety factors. Surely, it would need to be speculative, but that's OK.*

We will integrate this into the discussion.

> ***Reviewer #2 (Public Review):***
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> *Noticeably, the authors did not collect metabolic data nor did they model metabolic rates using their modelling framework. Instead, they performed a somewhat traditional inverse dynamics analysis from multiple animals hopping at a self-selected speed.*

We aimed to provide a joint-level explanation, but we will address the limitations of not modelling the energy consumers themselves (the skeletal muscles) in the revised manuscript. We plan to expand upon muscle level energetics in the future with a more detailed MSK model.

> *Within these analyses, the authors largely focused on ankle EMA, discussing its potential importance (because it affects tendon stress, which affects tendon strain energy, which*

> *affects muscle mechanics) on the metabolic cost of hopping. However, EMA was roughly estimated (CoP was fixed to the foot, not measured)...*

As noted in our methods, EMA was not calculated from a fixed centre of pressure (CoP). We did fix the medial-lateral position, owing to the fact that both feet contacted the force plate together, but the anteroposterior movement of the CoP was recorded by the force plate and thus allowed to move. We report the movement (or lack of movement) in our results. The anterior-posterior axis is the most relevant to lengthening or shortening the distance of the 'out-lever' R, and thereby EMA.

It is necessary to assume fixed medial-lateral position because a single force trace and CoP is recorded when two feet land on the force plate. The medial-lateral forces on each foot cancel out so there is no overall medial-lateral movement if the forces are symmetrical (e.g. if the kangaroo is hopping in a straight path and one foot is not in front of the other). We only used symmetrical trials so that the anterior-posterior movement of the CoP would be reliable.

> *and did not detectibly associate with hopping speed (see results).*
>
> *Yet, the authors interpret their EMA findings as though it systematically related with speed to explain their theory on how metabolic cost is unique in kangaroos vs. other animals.*

Indeed, the relationship between R and speed (and therefore EMA and speed) was not significant. However, the significant change in ankle height with speed, combined with no systematic change in COP at midstance, demonstrates that R would get longer at faster speeds. If we consider the nonsignificant relationship between R and speed to indicate that there is no change in R, then these two results conflict. We could not find a flaw in our methods, so instead concluded that the nonsignificant relationship between R and speed may be due to a small change in R being undetectable in our data. Taking both results into account, we think it is more likely that there is a non-detectable change in R, rather than no change in R with speed, but we presented both results for transparency.

> *These speed vs. biomechanics relationships were limited by comparisons across different animals hopping at different speeds and could have been strengthened using repeated measures design.*

There is significant variation in speed within individuals, not just between individuals. The preferred speed of kangaroos is 2-4.5 m/s, but most individuals show a wide range within this. Eight of our 16 kangaroos had a maximum speed that was between 1-2m/s faster than their slowest trial. Repeated measures of these eight individuals comprises 78 out of the 100 trials.

It would be ideal to collect data across the full range of speeds for all individuals, but it is not feasible in this type of experimental setting. Interference such as chasing is dangerous to kangaroos as they are prone to strong adverse reactions to stress.

> *There are also multiple inconsistencies between the authors' theory on how mechanics affect energetics and the cited literature, which leaves me somewhat confused and wanting more clarification and information on how mechanics and energetics relate.*

We will ensure that this is clearer in the revised manuscript.

> *My apologies for the less-than-favorable review, I think that this is a neat biomechanics study - but am unsure if it adds much to the literature on the topic of kangaroo hopping energetics in its current form.*

> ***Reviewer #3 (Public Review):***
>
> *Summary:*
>
> *The goal of this study is to understand how, unlike other mammals, kangaroos are able to increase hopping speed without a concomitant increase in metabolic cost. They use a biomechancial analysis of kangaroo hopping data across a range of speeds to investigate how posture, effective mechanical advantage, and tendon stress vary with speed and mass. The main finding is that a change in posture leads to increasing effective mechanical advantage with speed, which ultimately increases tendon elastic energy storage and returns via greater tendon strain. Thus kangaroos may be able to conserve energy with increasing speed by flexing more, which increases tendon strain.*
>
> *Strengths:*
>
> *The approach and effort invested into collecting this valuable dataset of kangaroo locomotion is impressive. The dataset alone is a valuable contribution.*

Thank you!

> *Weaknesses:*
>
> *Despite these strengths, I have concerns regarding the strength of the results and the overall clarity of the paper and methods used (which likely influences how convincingly the main results come across).*
>
> *(1) The paper seems to hinge on the finding that EMA decreases with increasing speed and that this contributes significantly to greater tendon strain estimated with increasing speed. It is very difficult to be convinced by this result for a number of reasons:*
>
> *• It appears that kangaroos hopped at their preferred speed. Thus the variability observed is across individuals not within. Is this large enough of a range (either within or across subjects) to make conclusions about the effect of speed, without results being susceptible to differences between subjects?*

Apologies, this was not clear in the manuscript. Kangaroos hopping at their preferred speed means we did not chase or startle them into high speeds to comply with ethics and enclosure limitations. Thus we did not record a wide range of speed within the bounds of what kangaroos are capable of (up to 12 m/s), but for the range we did measure (~2-4.5 m/s), there is variation hopping speed within each individual kangaroo. Out of 16 individuals, eight individuals had a difference of 1-2m/s between their slowest and fastest trials, and these kangaroos accounted for 78 out of 100 trials. Of the remainder, six individuals had three for fewer trials each, and two individual had highly repeatable speeds (3 out of 4, and 6 out of 7 trials were within 0.5 m/s). We will ensure this is clear in the revised manuscript.

> *In the literature cited, what was the range of speeds measured, and was it within or between subjects?*

For other literature, to our knowledge the highest speed measured is ~9.5m/s (see supplementary Fig1b) and there were multiple measures for several individuals (see methods Kram & Dawson 1998).

> *• Assuming that there is a compelling relationship between EMA and velocity, how reasonable is it to extrapolate to the conclusion that this increases tendon strain and ultimately saves metabolic cost?*

> *They correlate EMA with tendon strain, but this would still not suggest a causal relationship (incidentally the p-value for the correlation is not reported).*

We will add supporting literature on the relationship between metabolic cost and tendon stress (or strain), to elaborate on why the correlation between EMA and stress is important.

> *Tendon strain could be increasing with ground reaction force, independent of EMA.*
>
> *Even if there is a correlation between strain and EMA, is it not a mathematical necessity in their model that all else being equal, tendon stress will increase as ema decreases? I may be missing something, but nonetheless, it would be helpful for the authors to clarify the strength of the evidence supporting their conclusions.*

Yes, GRF also contributes to the increase in tendon stress in the mechanism we propose. We have illustrated this in Fig 6, however we will make this clearer in the revised discussion.

> *• The statistical approach is not well-described. It is not clear what the form of the statistical model used was and whether the analysis treated each trial individually or grouped trials by the kangaroo. There is also no mention of how many trials per kangaroo, or the range of speeds (or masses) tested.*

The methods include the statistical model with the variables that we used, as well as the kangaroo masses (13.7 to 26.6 kg, mean: 20.9 ± 3.4 kg). We will move the range of speeds from the supplementary material to the results or figure captions. We will add information on the number of trials per kangaroo to the methods.

We did not group the data e.g. by using an average speed per individual for all their trials, or by comparing fast to slow groups (this was for display purposes in our figures, which we will make clearer in the methods).

> *Related to this, there is no mention of how different speeds were obtained. It seems that kangaroos hopped at a self-selected pace, thus it appears that not much variation was observed. I appreciate the difficulty of conducting these experiments in a controlled manner, but this doesn't exempt the authors from providing the details of their approach.*
>
> *• Some figures (Figure 2 for example) present means for one of three speeds, yet the speeds are not reported (except in the legend) nor how these bins were determined, nor how many trials or kangaroos fit in each bin. A similar comment applies to the mass categories. It would be more convincing if the authors plotted the main metrics vs. speed to illustrate the significant trends they are reporting.*

Thank you for this comment. The bins are used only for display purposes and not within the analysis. In the revised manuscript, we will ensure this is clear.

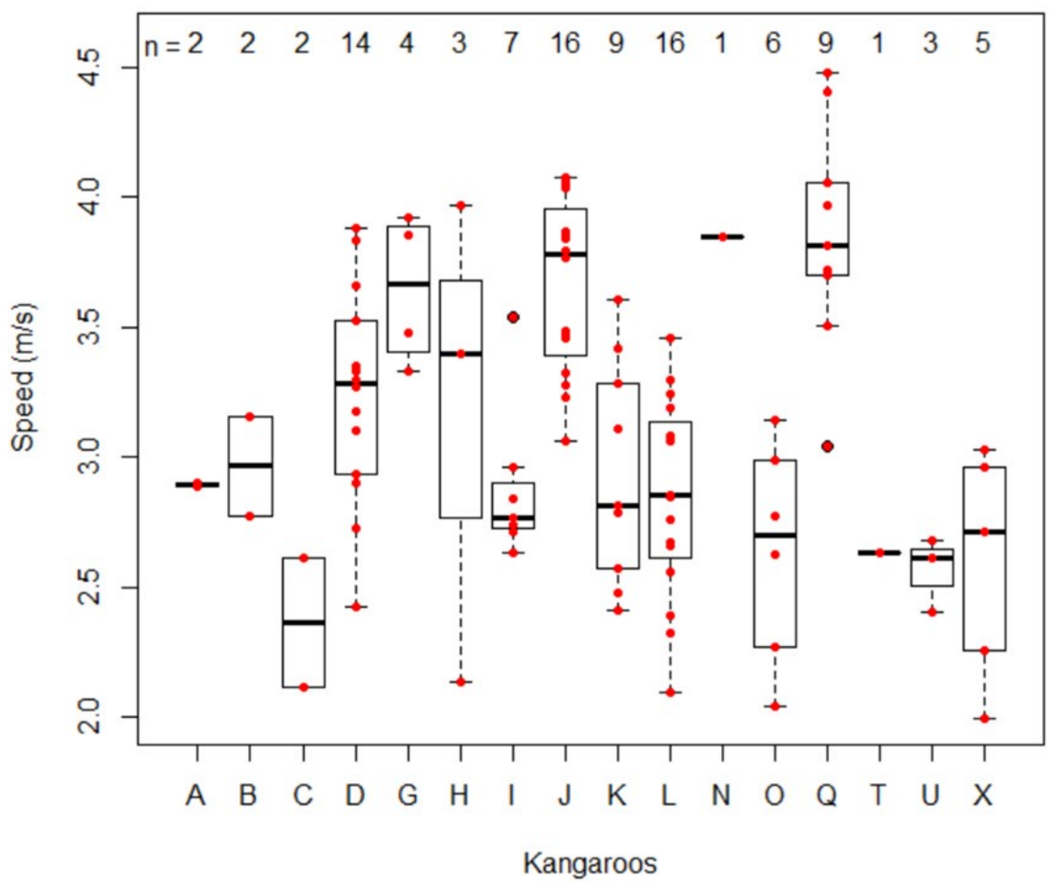
> *(2) The significance of the effects of mass is not clear. The introduction and abstract suggest that the paper is focused on the effect of speed, yet the effects of mass are reported throughout as well, without a clear understanding of the significance. This weakness is further exaggerated by the fact that the details of the subject masses are not reported.*

Indeed, the primary aim of our study was to explore the influence of speed, given the uncoupling of energy from hopping speed in kangaroos. We included mass to ensure that the effects of speed were not driven by body mass (i.e.: that larger kangaroos hopped faster).

(3) The paper needs to be significantly re-written to better incorporate the methods into the results section. Since the results come before the methods, some of the methods must necessarily be described such that the study can be understood at some level without turning to the dedicated methods section. As written, it is very difficult to understand the basis of the approach, analysis, and metrics without turning to the methods.

We agree, and in the revised manuscript will incorporate some of the methodological details within the results.

Author response image 1.



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